

Simple Regression:

Linear regression with one input

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Recall Task:

Predicting house prices

How much is my house worth?



How much is my house worth?



Look at recent sales in my neighborhood

- How much did they sell for?



Regression fundamentals: data, model, task

Data



input *output*
 $(x_1 = \text{sq.ft.}, y_1 = \$)$



$(x_2 = \text{sq.ft.}, y_2 = \$)$



$(x_3 = \text{sq.ft.}, y_3 = \$)$



$(x_4 = \text{sq.ft.}, y_4 = \$)$



$(x_5 = \text{sq.ft.}, y_5 = \$)$

⋮

Data



input *output*
($x_1 = \text{sq.ft.}$, $y_1 = \$$)



($x_2 = \text{sq.ft.}$, $y_2 = \$$)



($x_3 = \text{sq.ft.}$, $y_3 = \$$)



($x_4 = \text{sq.ft.}$, $y_4 = \$$)



($x_5 = \text{sq.ft.}$, $y_5 = \$$)

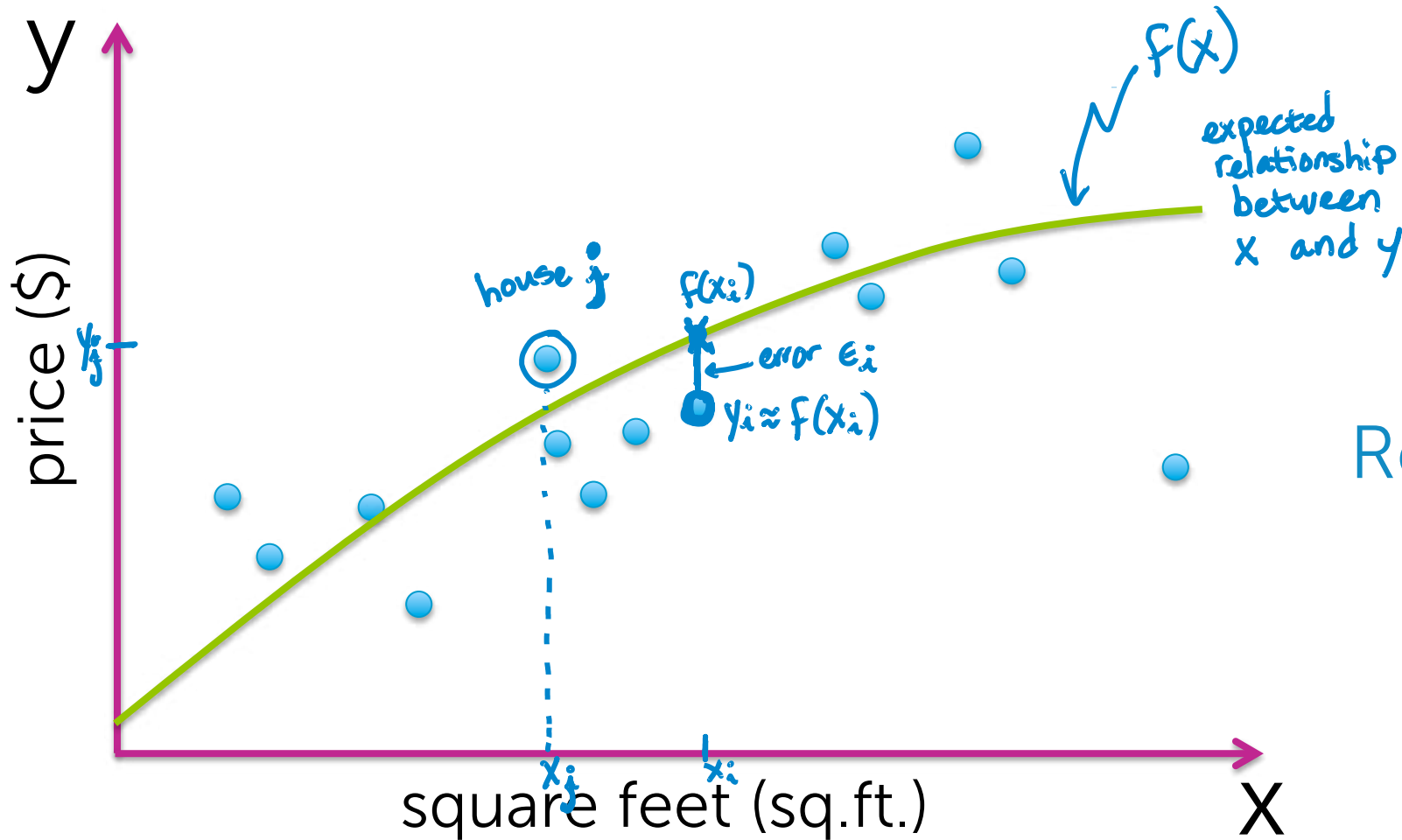
⋮

Input vs. Output:

- y is the quantity of interest
- assume y can be predicted from x

Model –

How we *assume* the world works



Regression model:

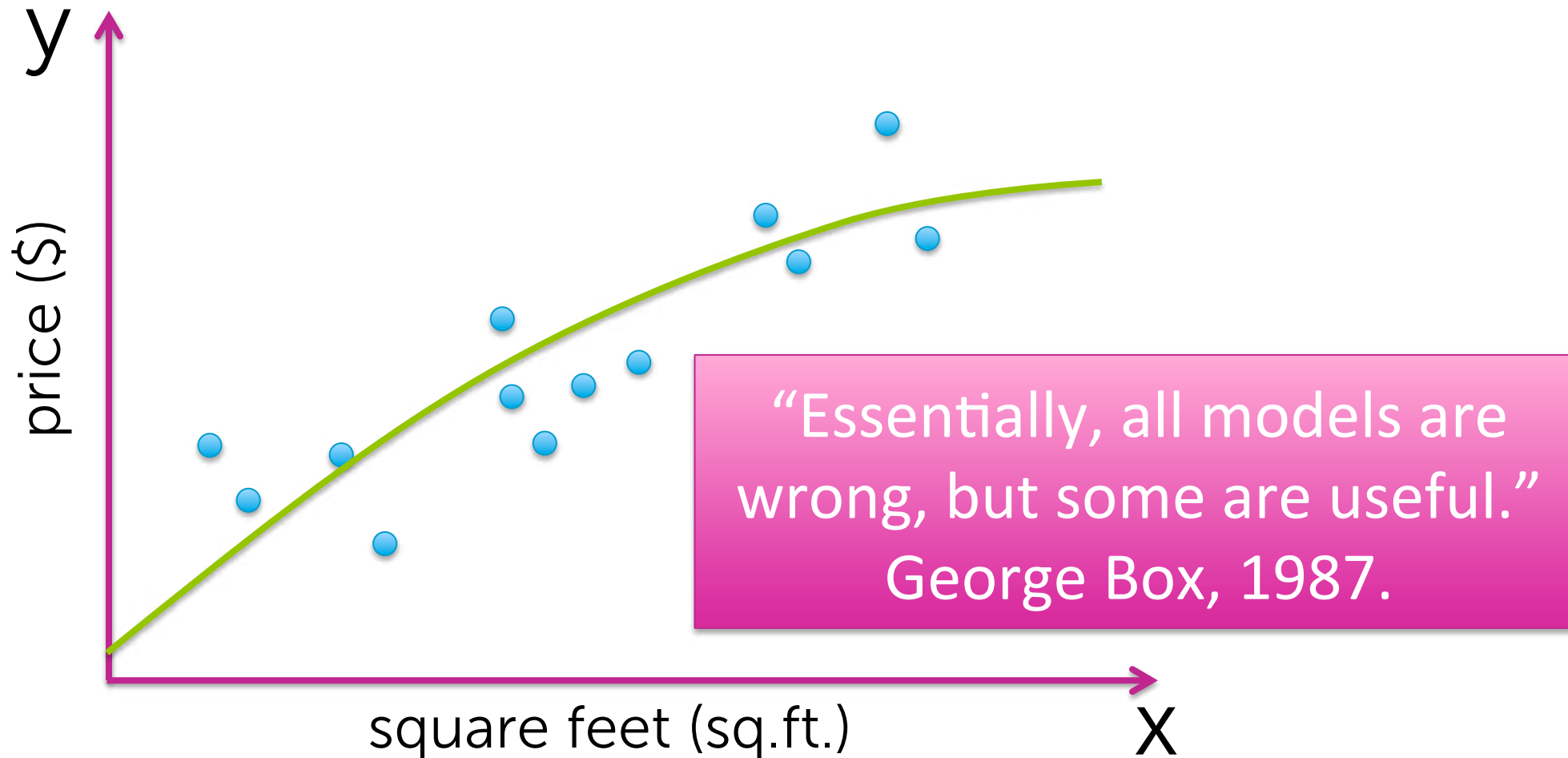
$$y_i = f(x_i) + \epsilon_i$$

$E[\epsilon_i] = 0$ ← equally likely that error is + or -
↑ expected value

⇓
 y_i is equally likely to be above or below $f(x_i)$

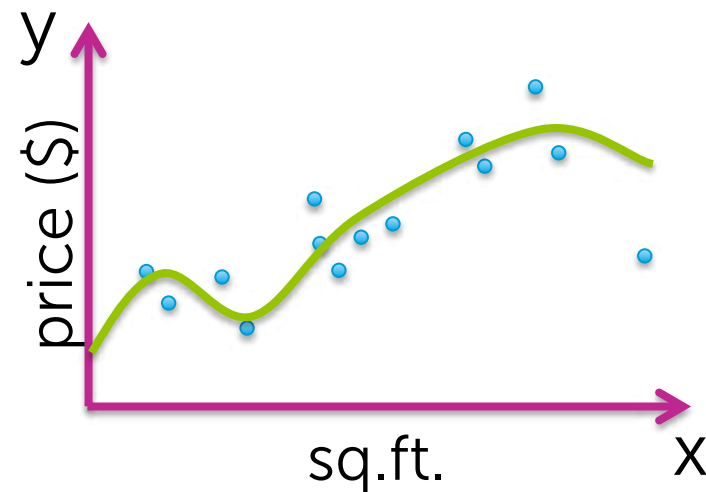
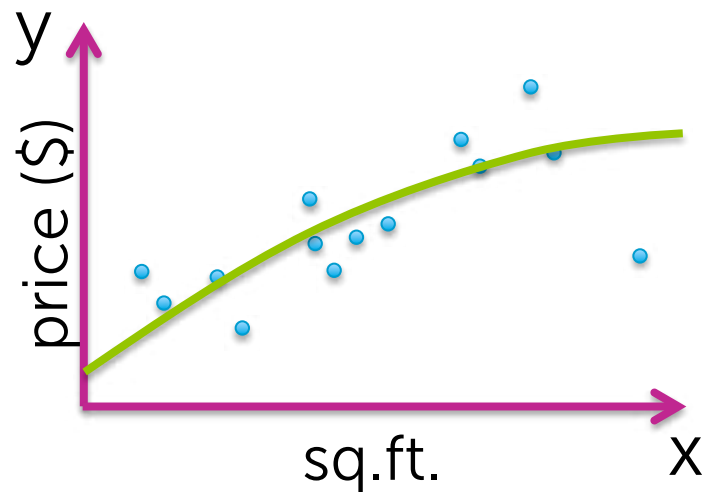
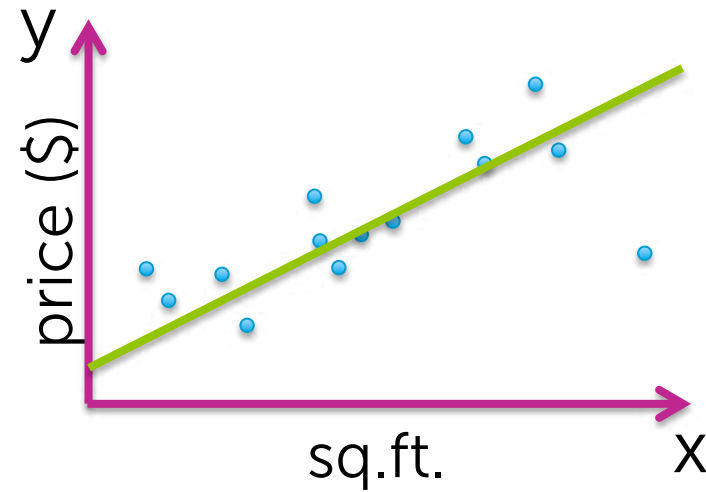
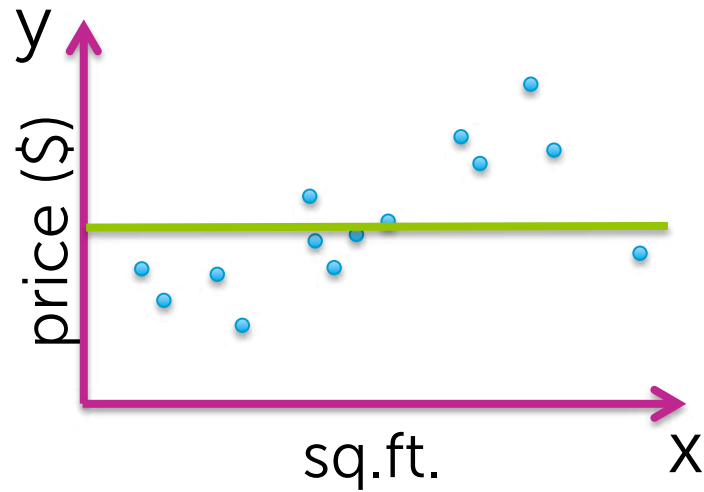
Model –

How we *assume* the world works

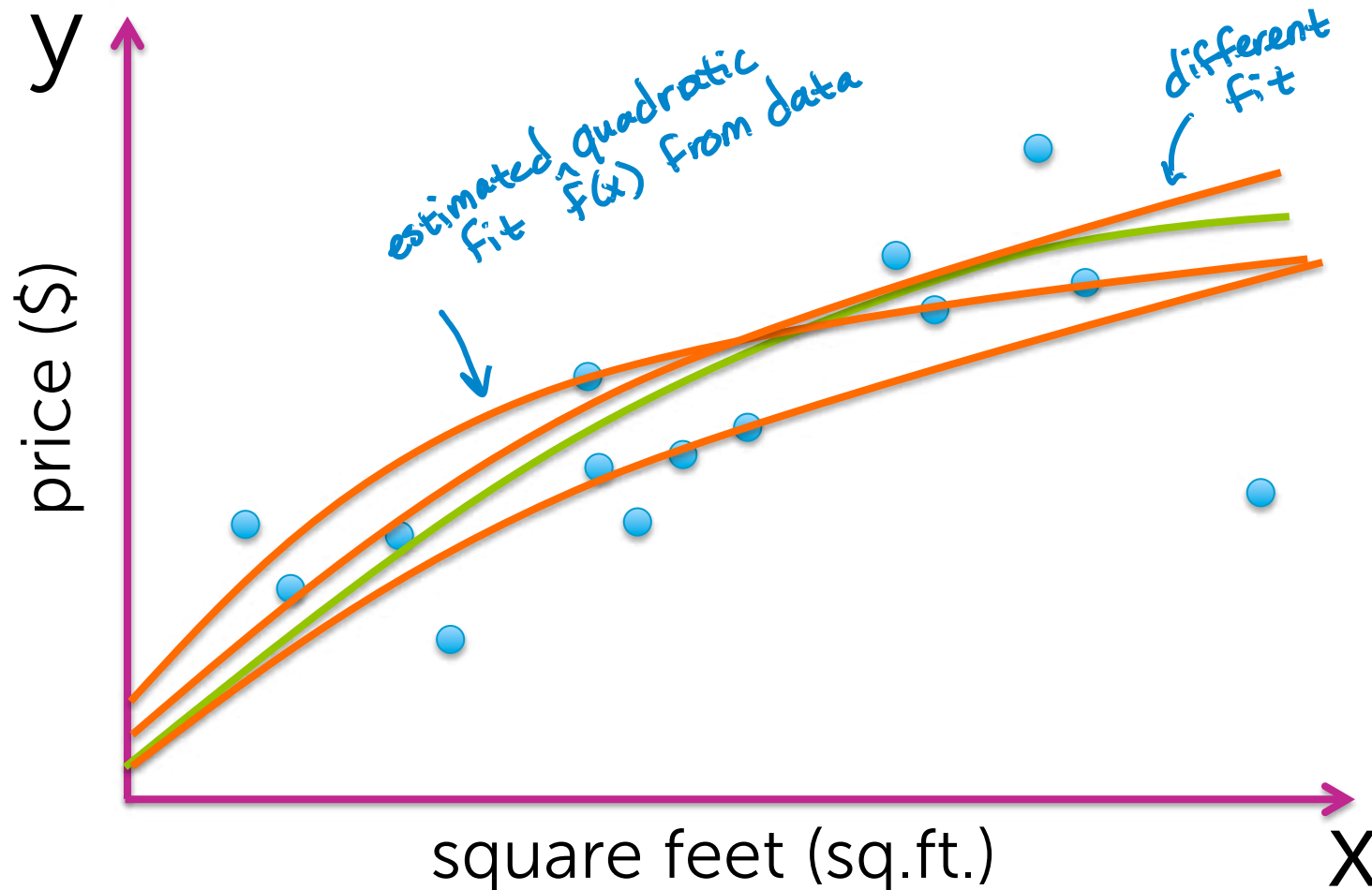


Task 1–

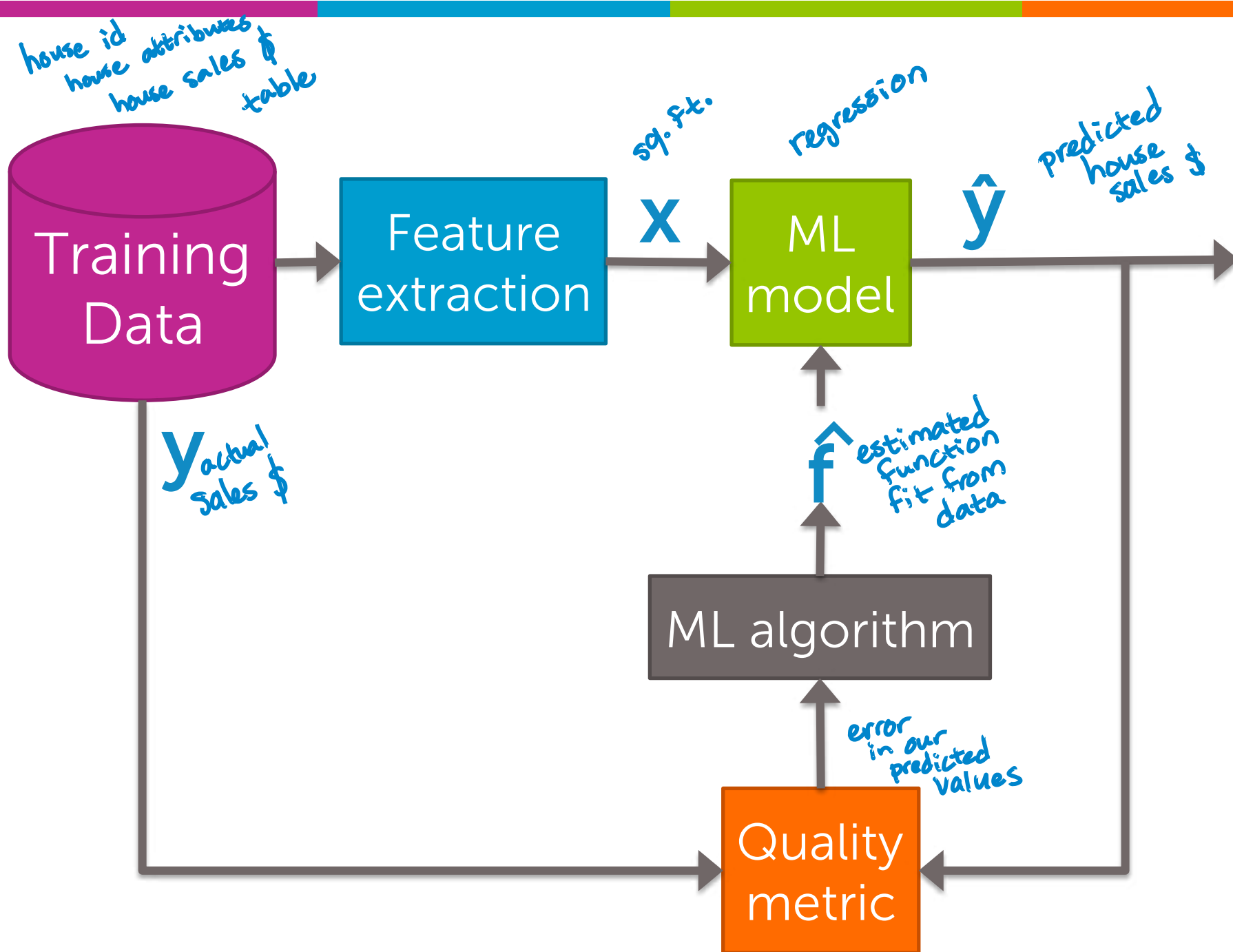
Which model $f(x)$?



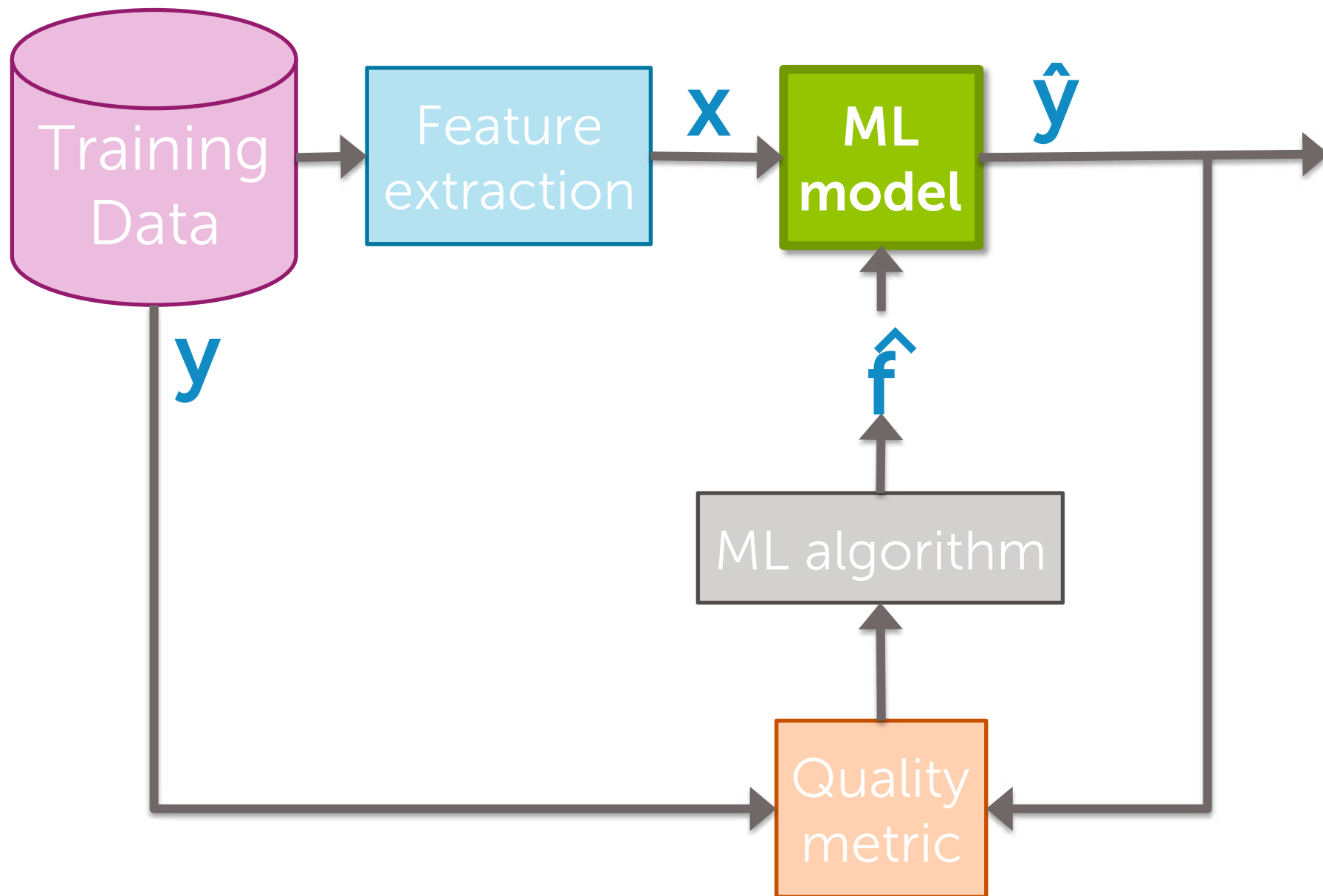
Task 2 – For a given model $f(x)$, estimate function $\hat{f}(x)$ from data



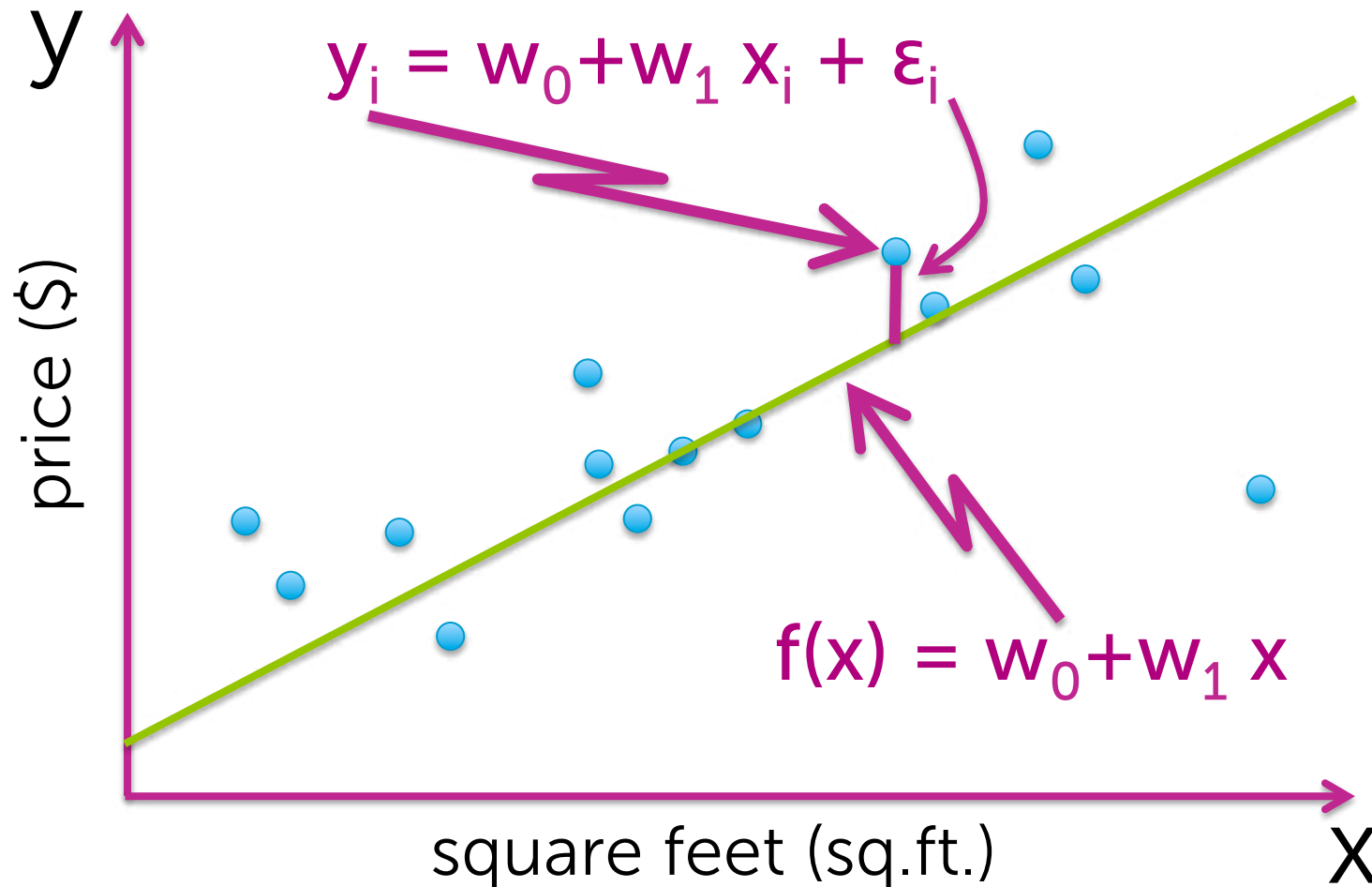
Assume model $f(x)$ is a quadratic function



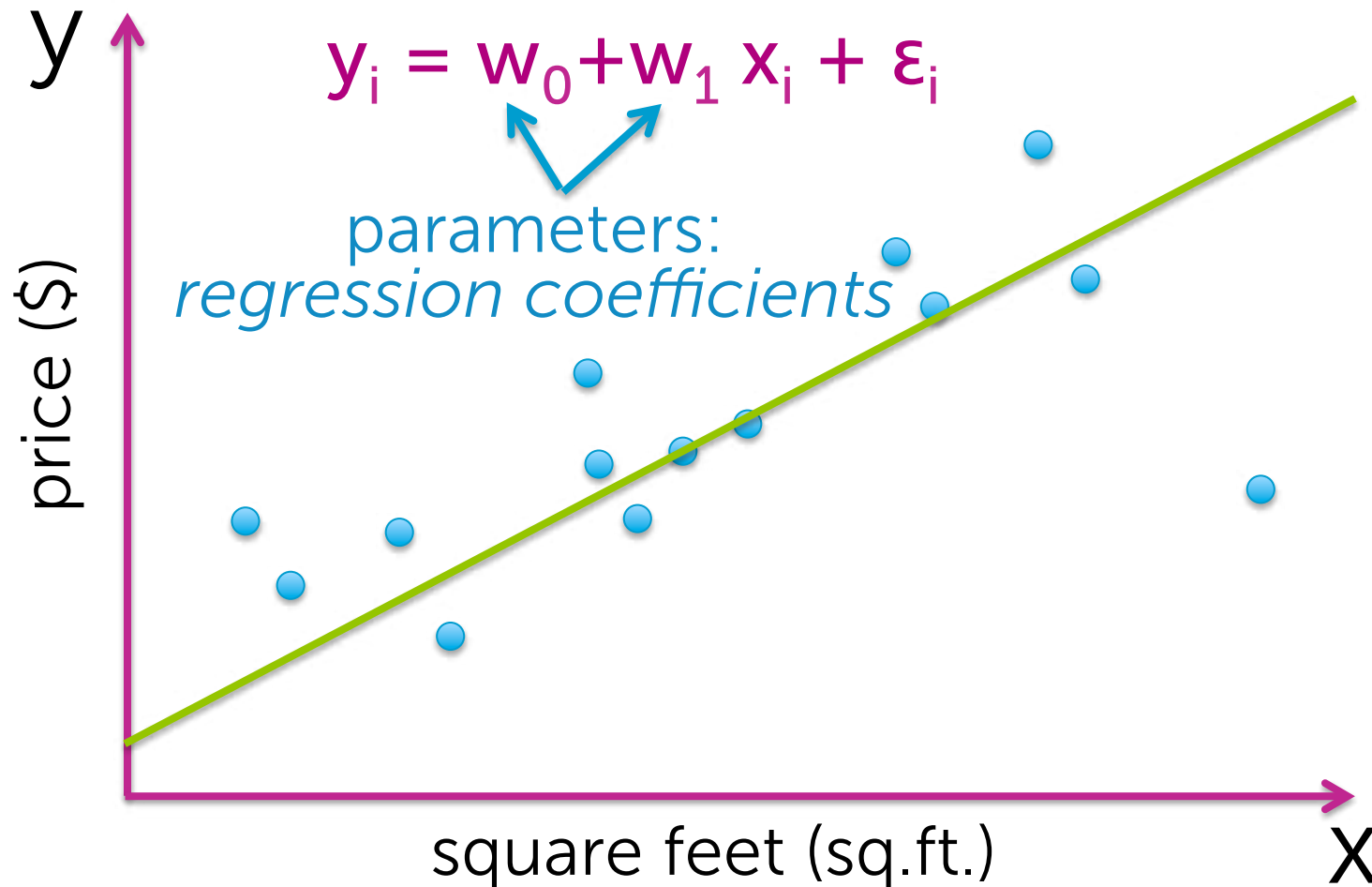
Simple linear regression



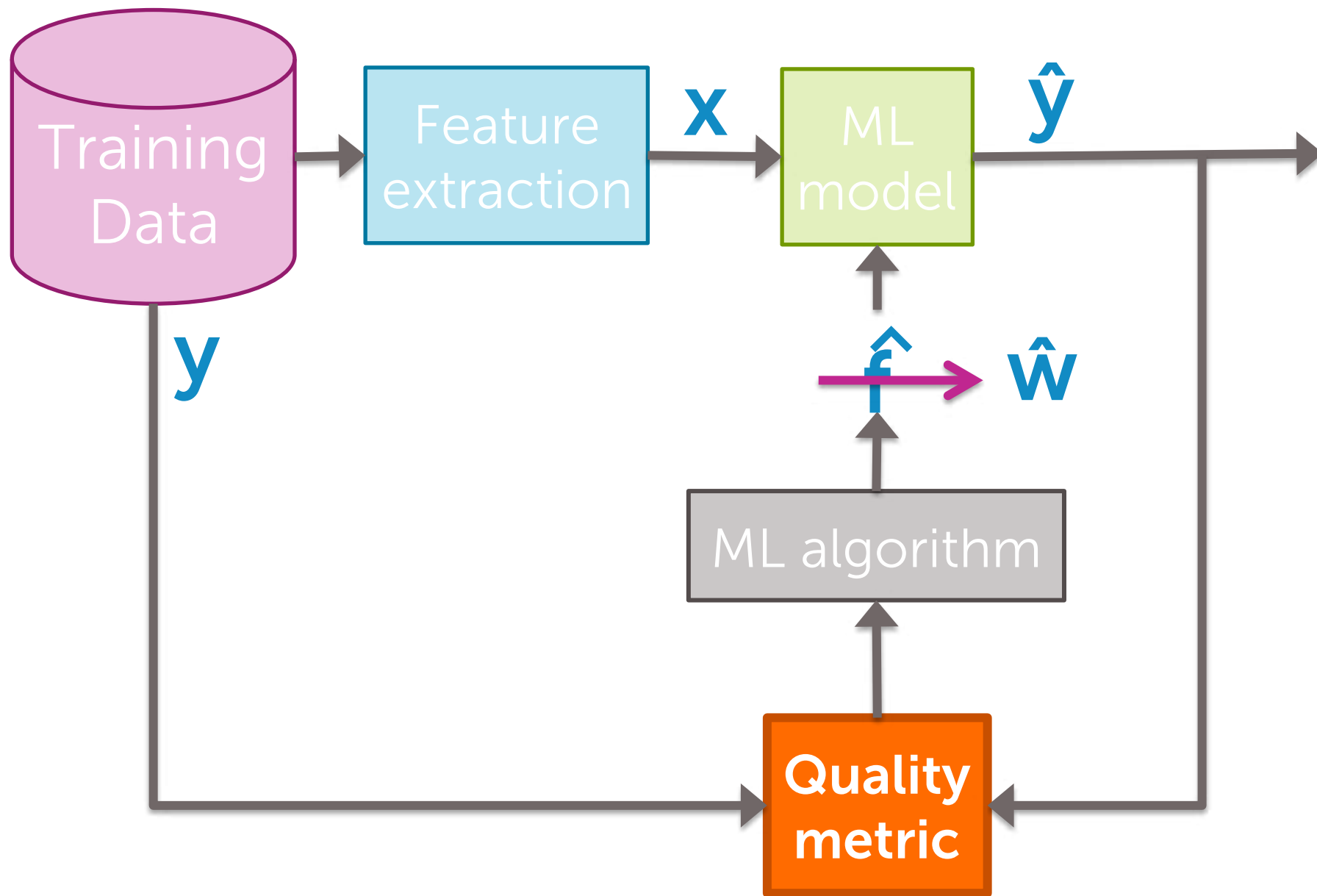
Simple linear regression model



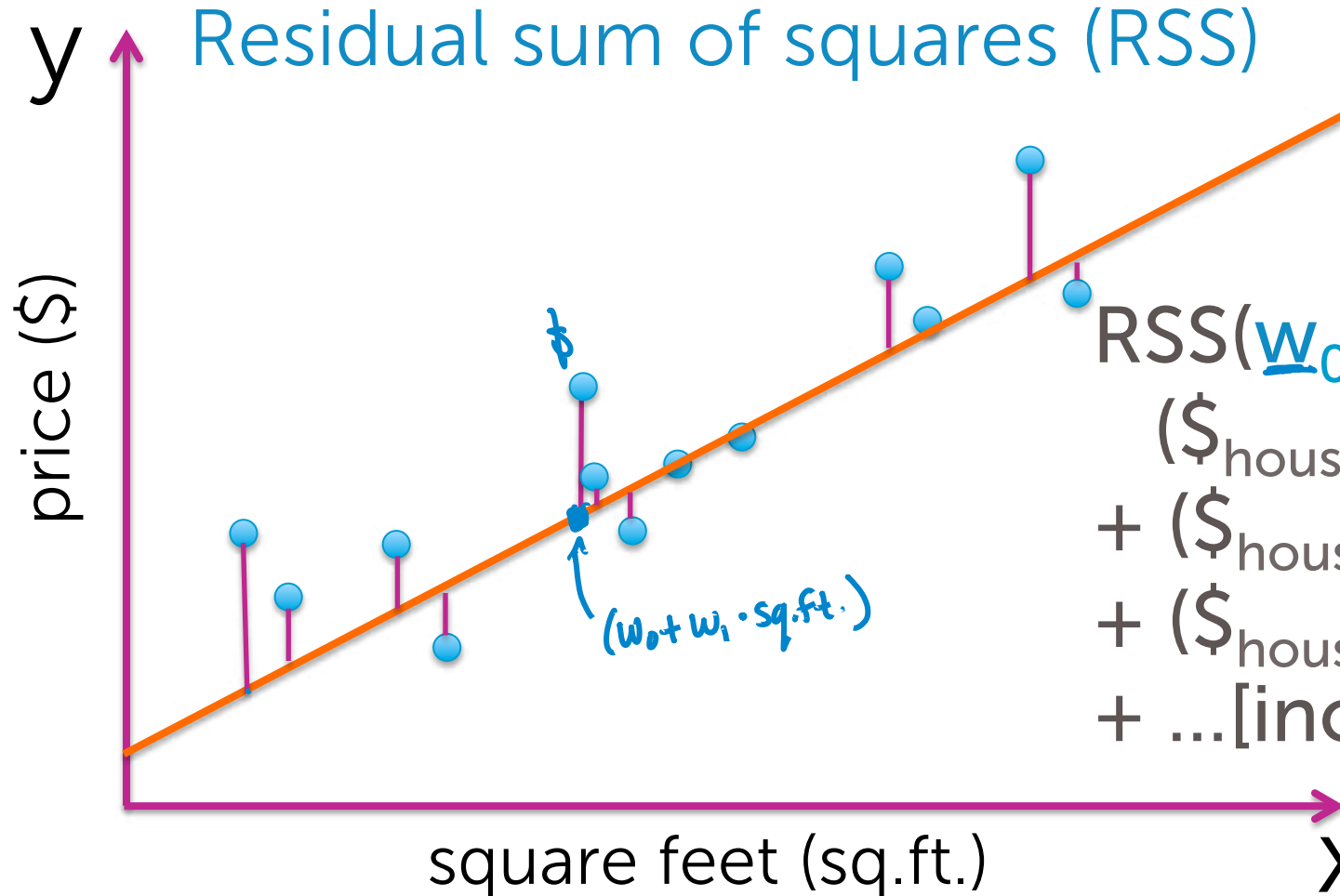
Simple linear regression model



Fitting a line to data

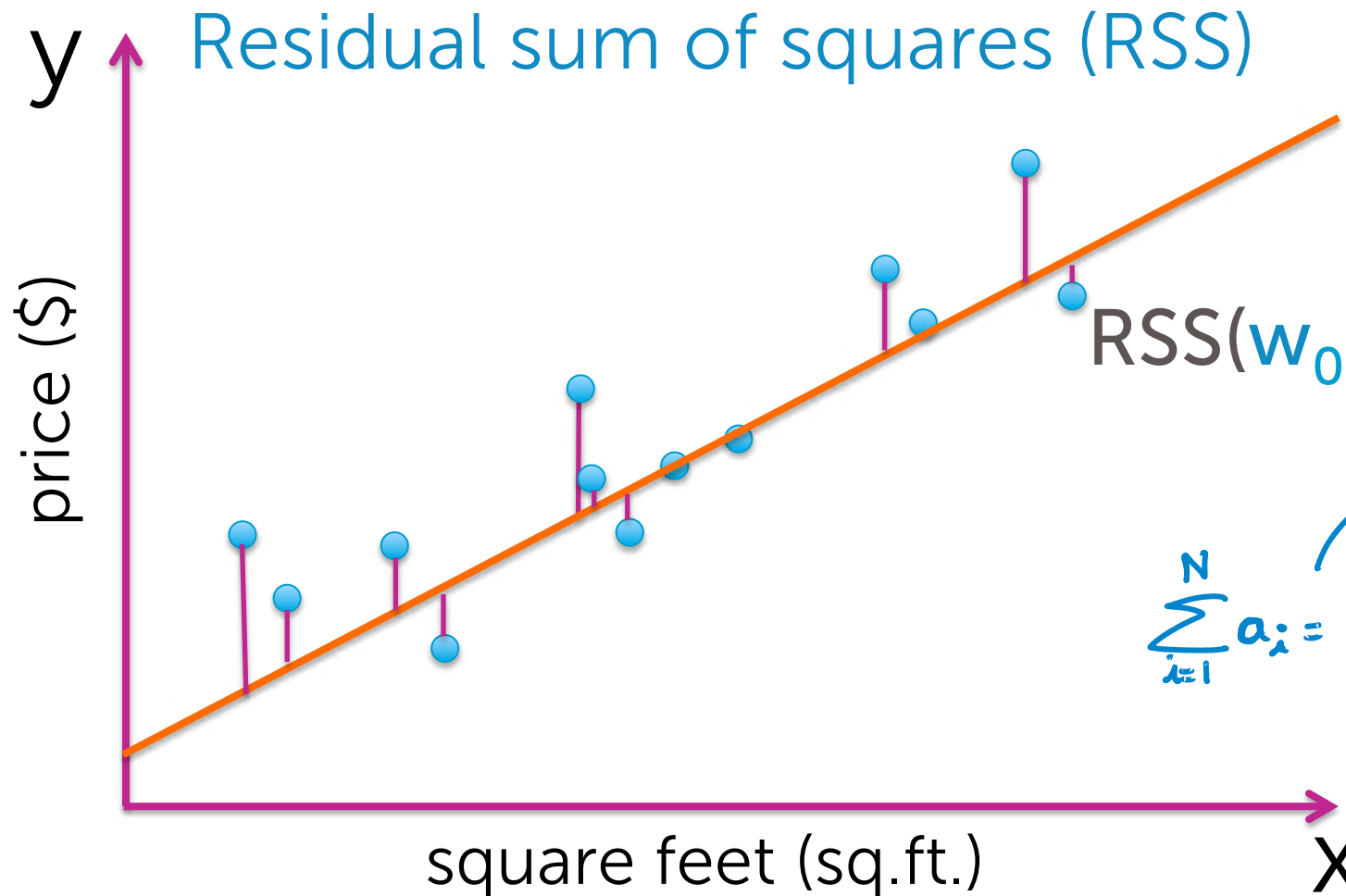


"Cost" of using a given line



$$\begin{aligned} \text{RSS}(\underline{w}_0, \underline{w}_1) = & (\$_{\text{house 1}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 1}}])^2 \\ & + (\$_{\text{house 2}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 2}}])^2 \\ & + (\$_{\text{house 3}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 3}}])^2 \\ & + \dots [\text{include all training houses}] \end{aligned}$$

"Cost" of using a given line



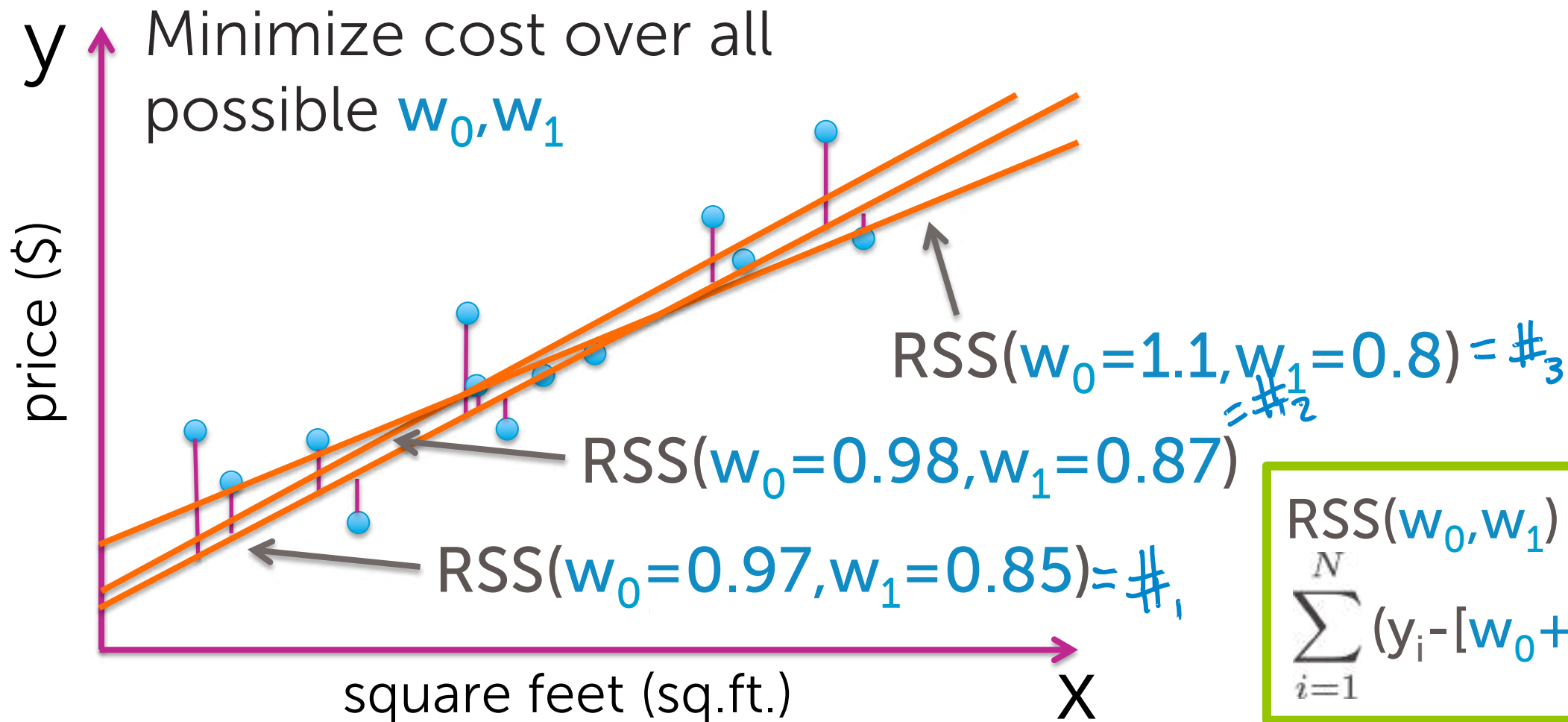
$$\text{RSS}(w_0, w_1) = \sum_{i=1}^N (y_i - [w_0 + w_1 x_i])^2$$

$$\sum_{i=1}^N a_i = a_1 + a_2 + \dots + a_N$$

Here,

$$a_i = (y_i - [w_0 + w_1 x_i])^2$$

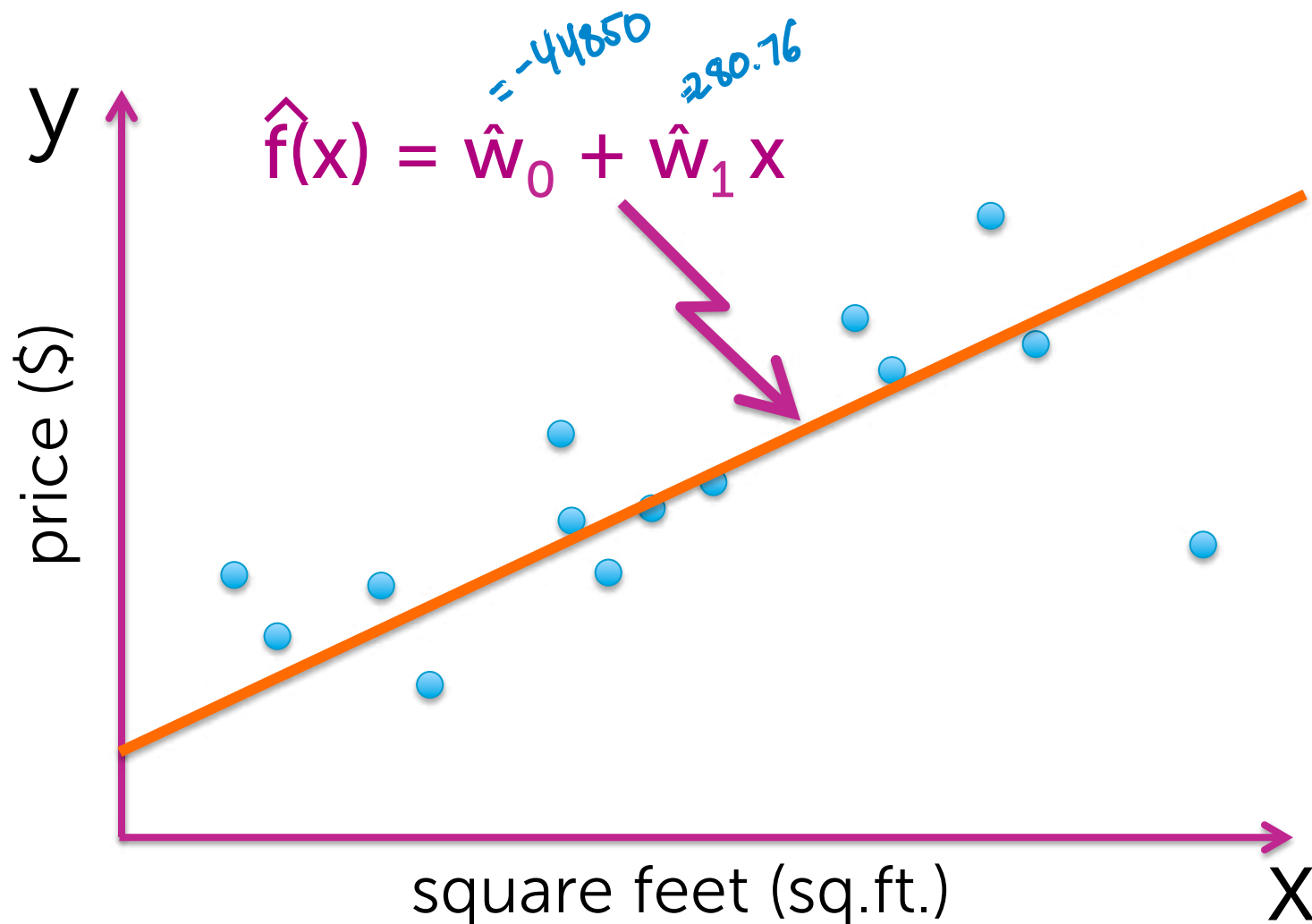
Find "best" line



$$RSS(w_0, w_1) = \sum_{i=1}^N (y_i - [w_0 + w_1 x_i])^2$$

The fitted line: use + interpretation

Model vs. fitted line



Regression model:

$$y_i = w_0 + w_1 x_i + \epsilon_i$$

parameters (unknown variables)

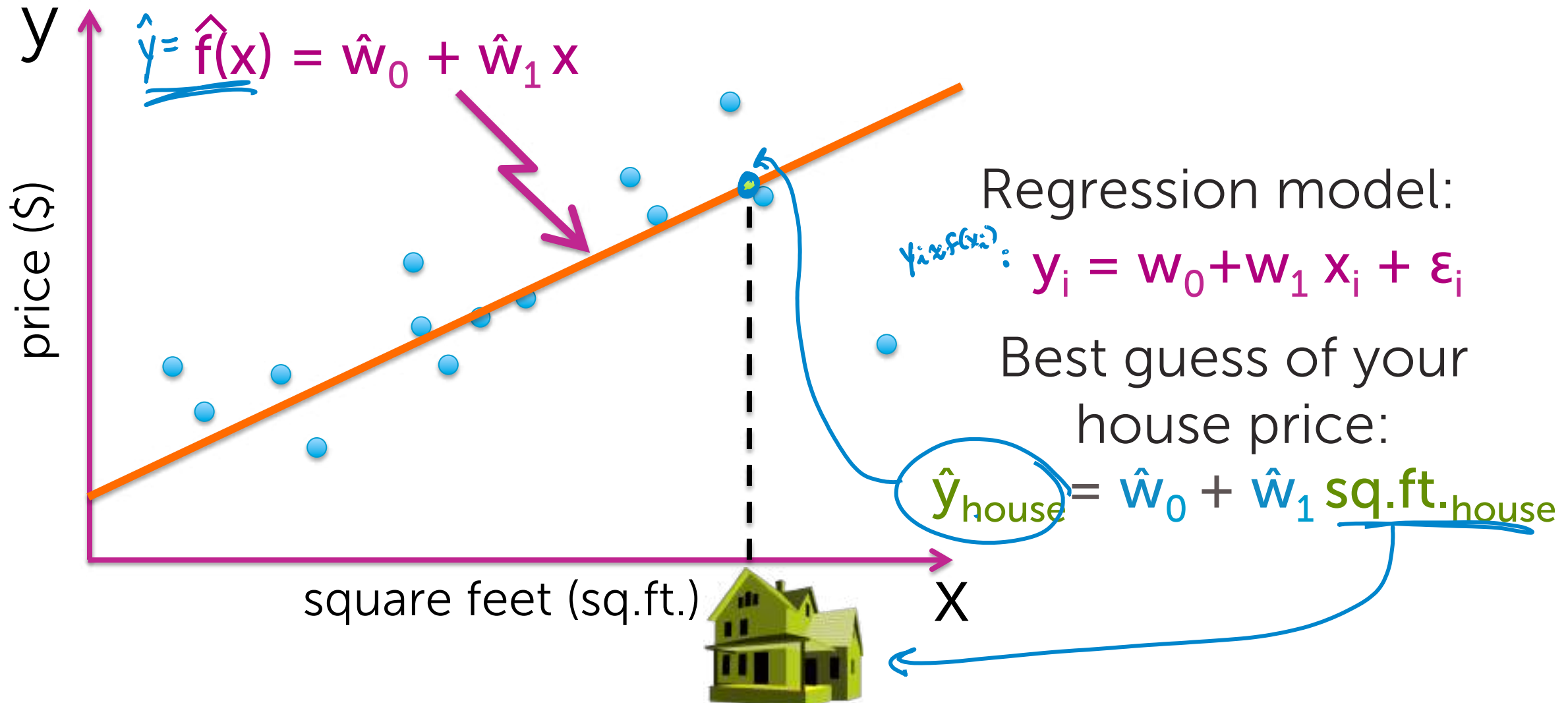
Estimated parameters:

$$\hat{w}_0, \hat{w}_1$$

take actual values

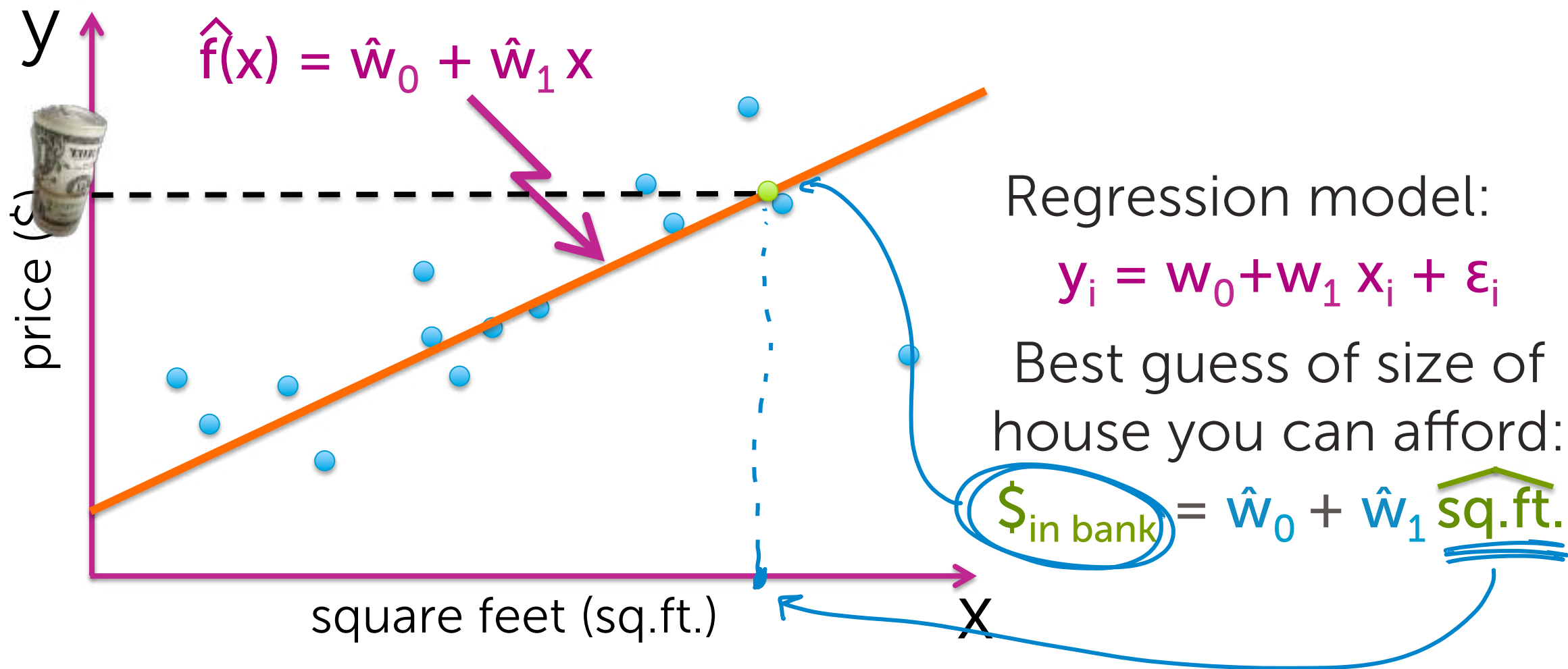
Seller:

Predicting your house price

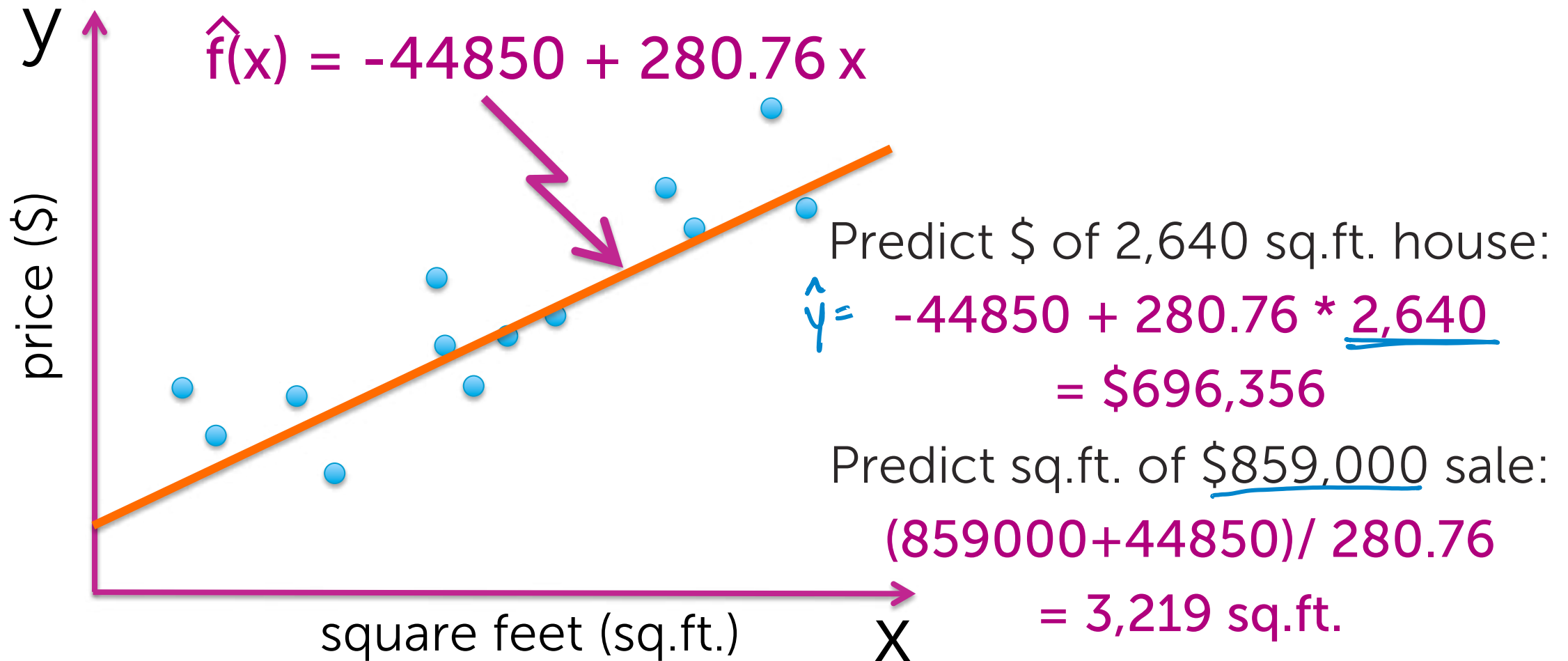


Buyer:

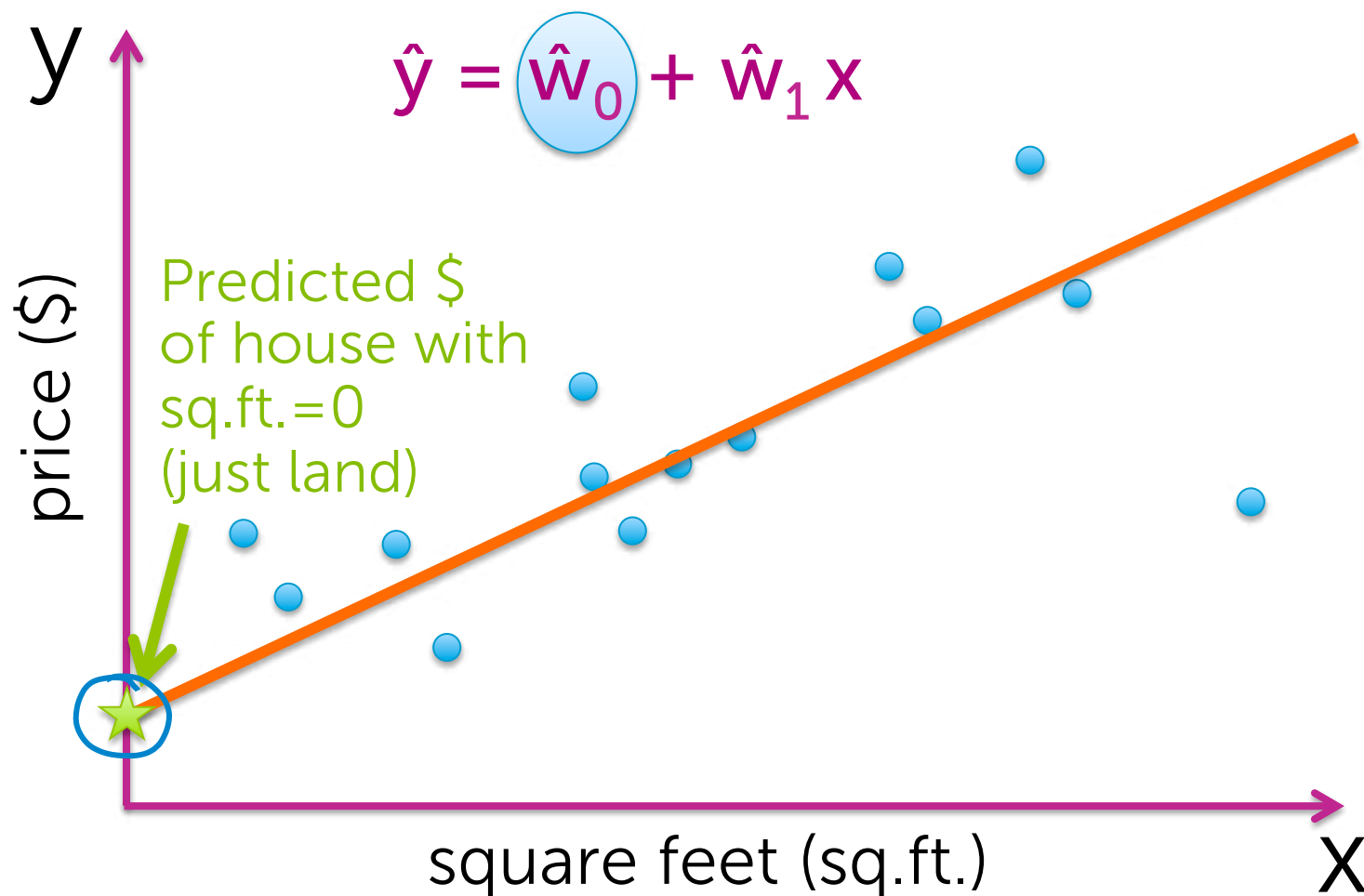
Predicting size of house



A concrete example



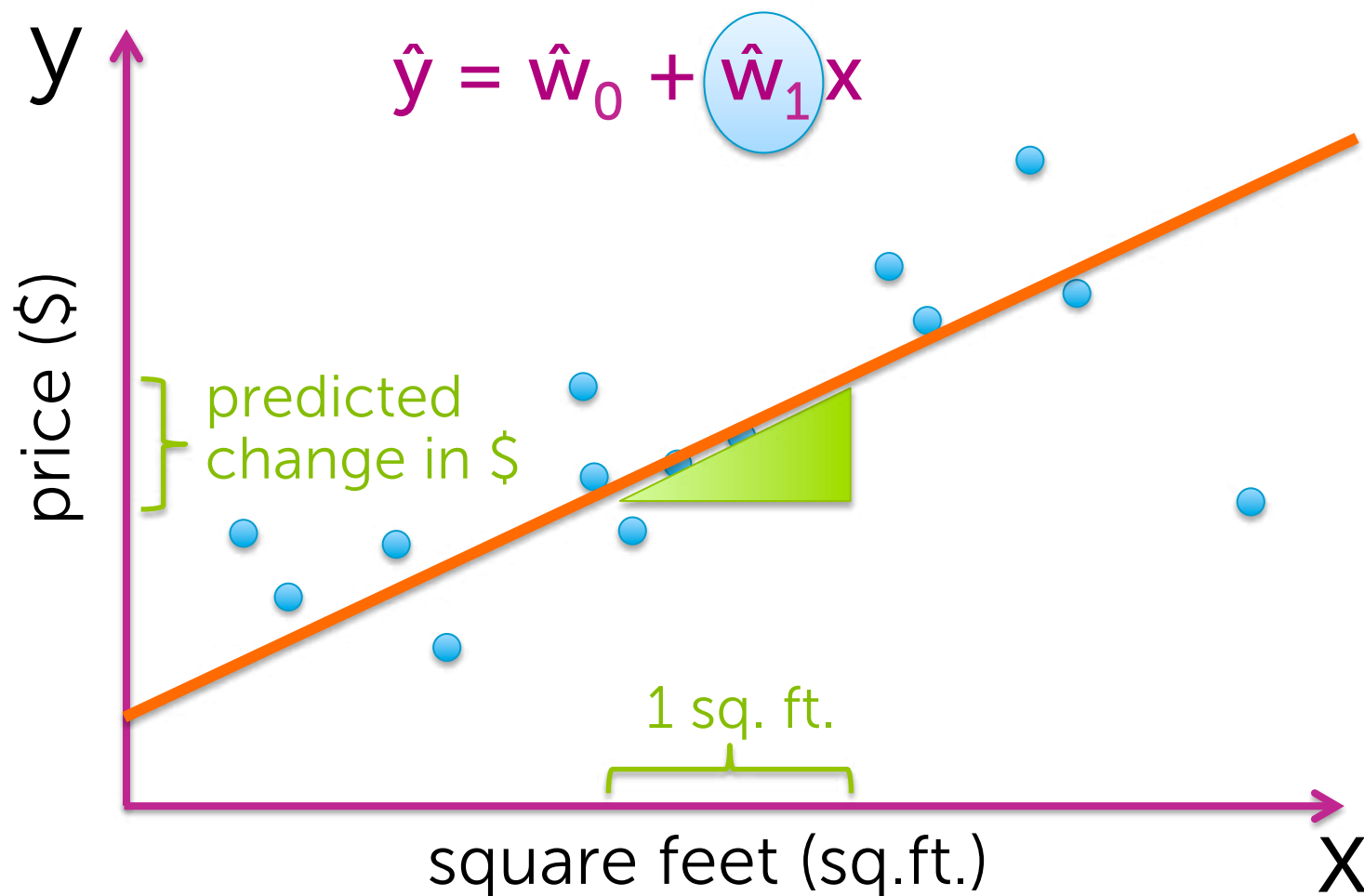
Interpreting the coefficients



$$\hat{y} = \hat{w}_0 \text{ when } x=0$$

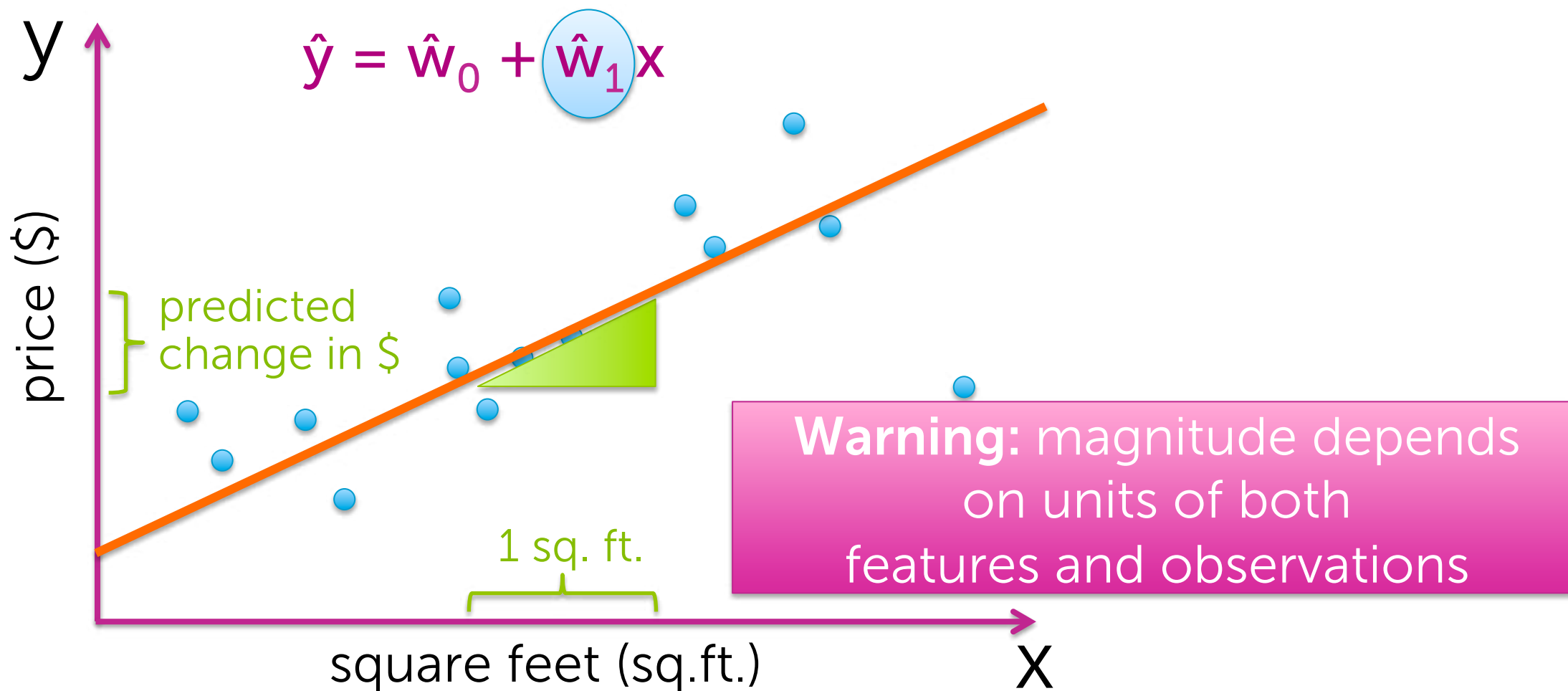
not very meaningful

Interpreting the coefficients

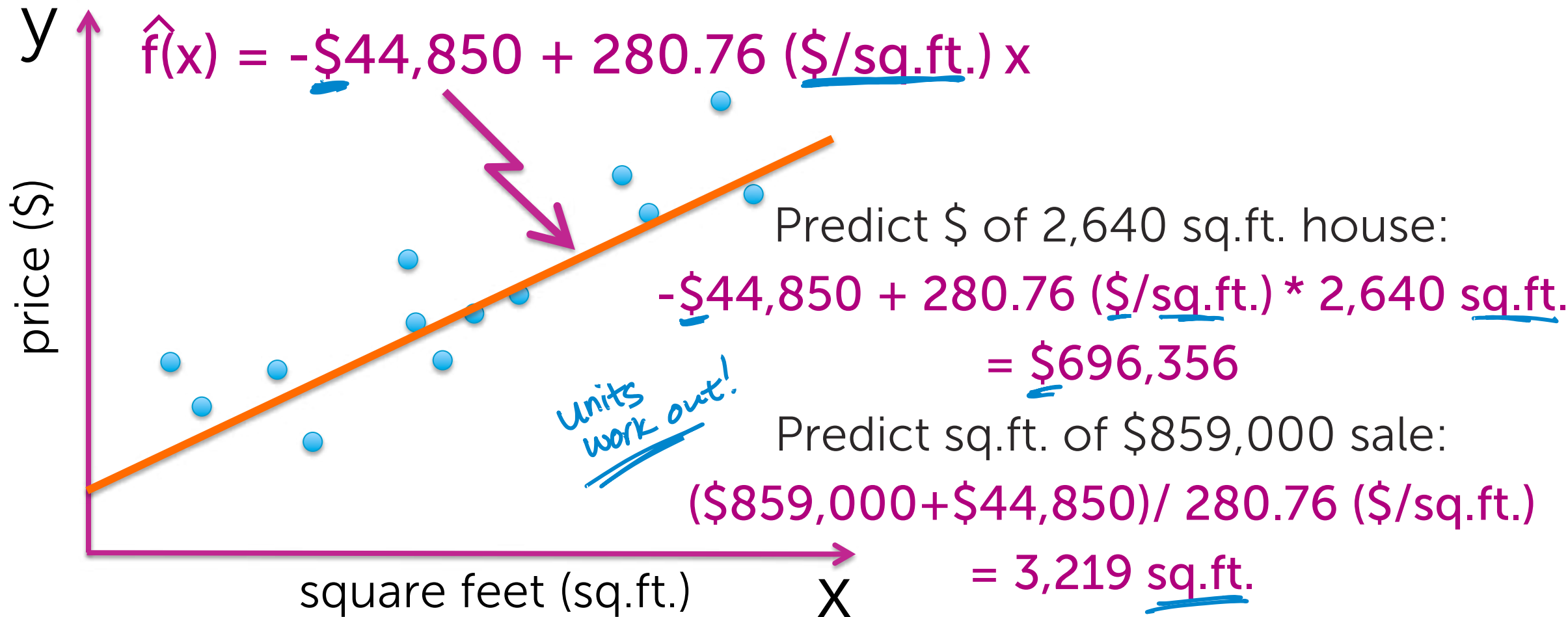


$$\begin{aligned} & \hat{\$}_{1001 \text{ sq.ft.}} - \hat{\$}_{1000 \text{ sq.ft.}} \\ &= \cancel{\hat{w}_0} + \hat{w}_1 \cdot 1001 \text{ sq.ft.} \\ &\quad - (\cancel{\hat{w}_0} + \hat{w}_1 \cdot 1000 \text{ sq.ft.}) \\ &= \hat{w}_1 \\ &\text{predicted change in the output} \\ &\quad \text{per unit change in input} \end{aligned}$$

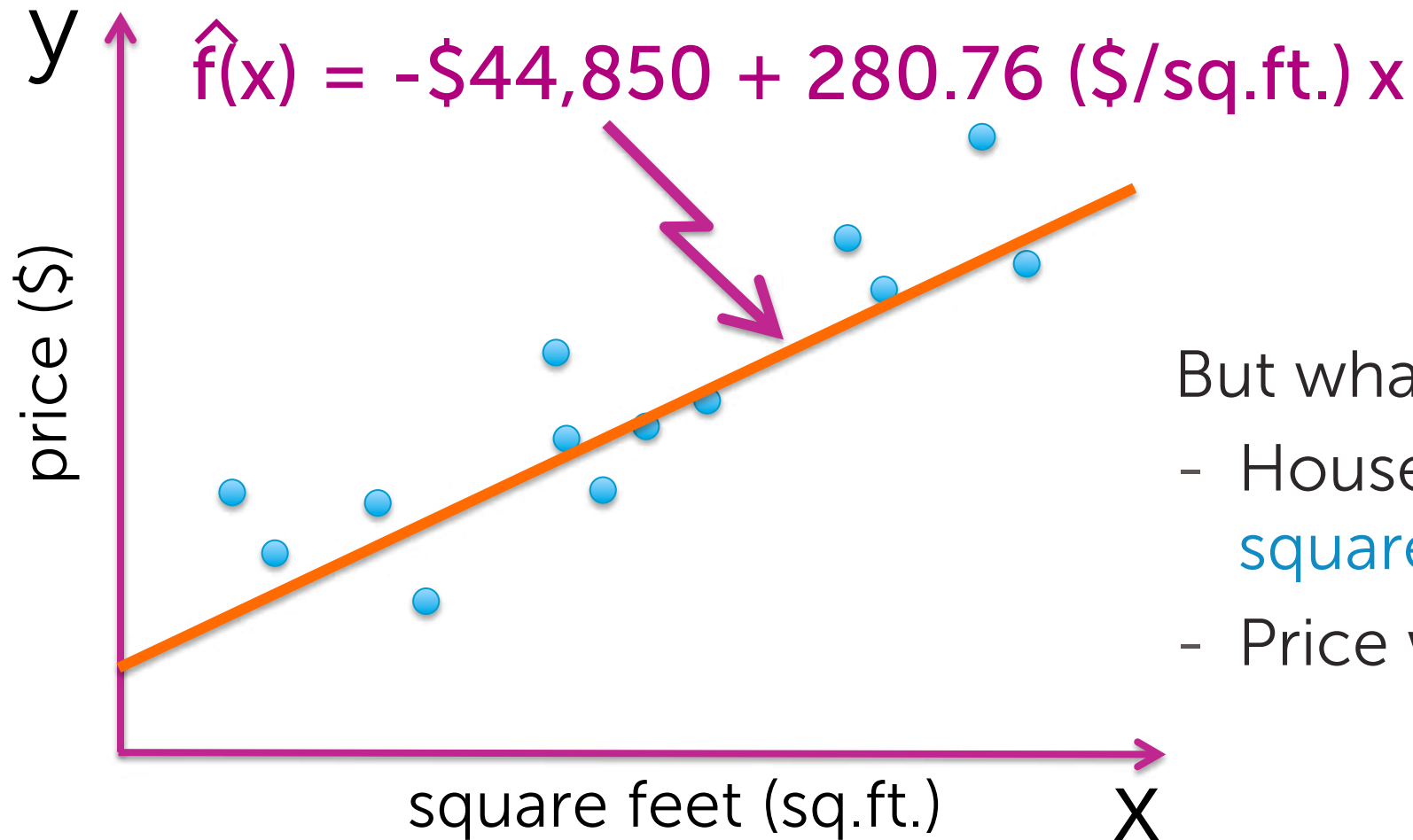
Interpreting the coefficients



A concrete example



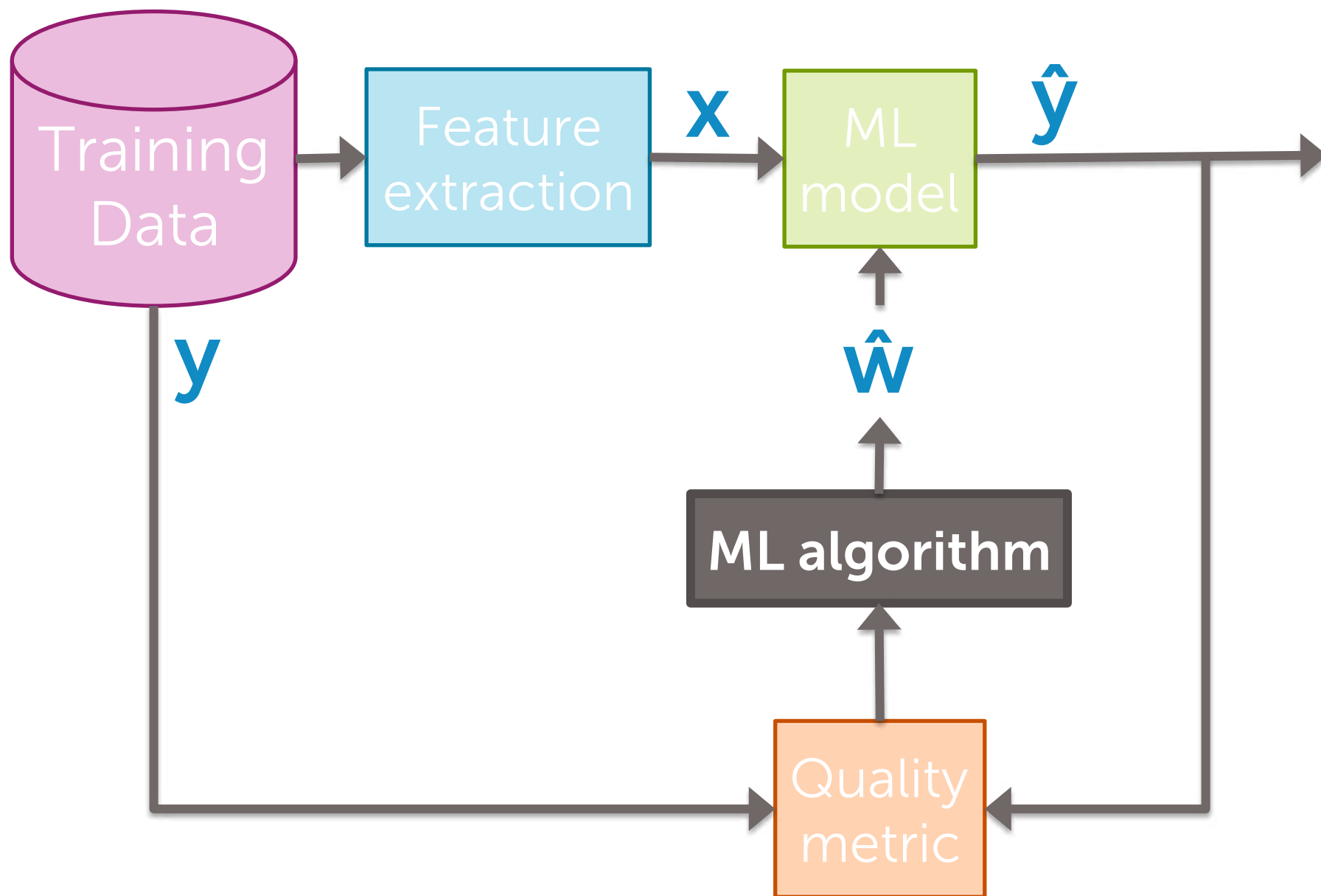
A concrete example



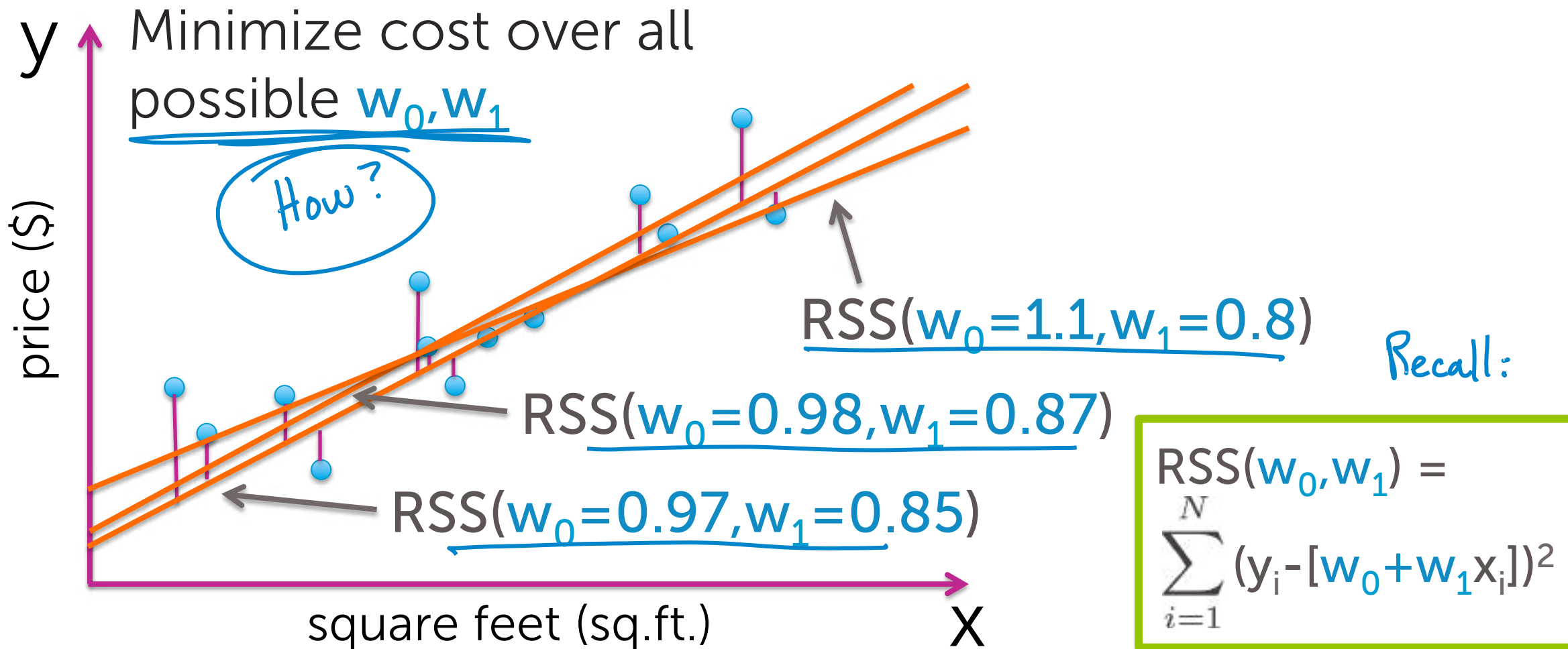
But what if:

- House was measured in **square meters**?
- Price was measured in **RMB**?

Algorithms for fitting the model

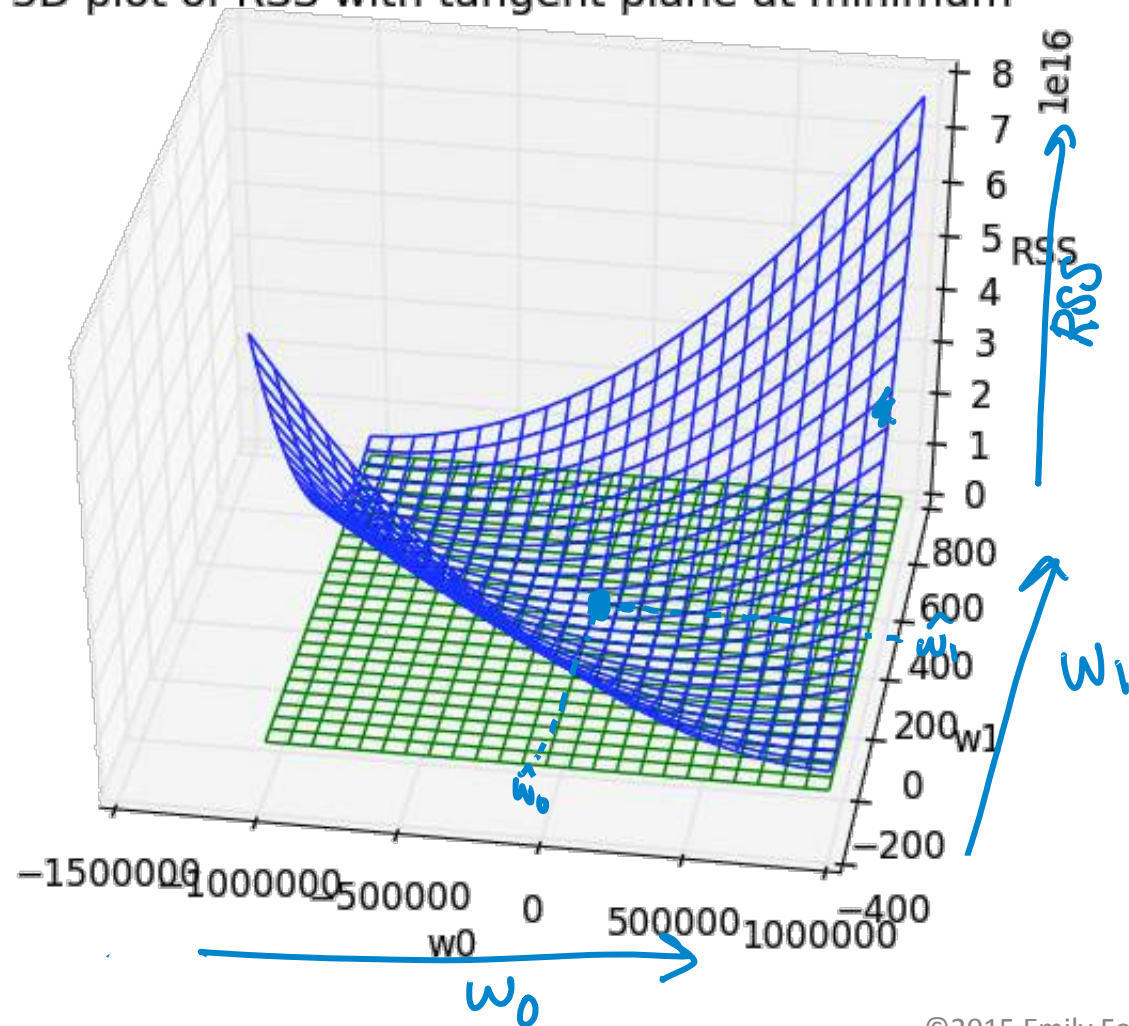


Find "best" line



Minimizing the cost

3D plot of RSS with tangent plane at minimum



Minimize function
over all possible w_0, w_1

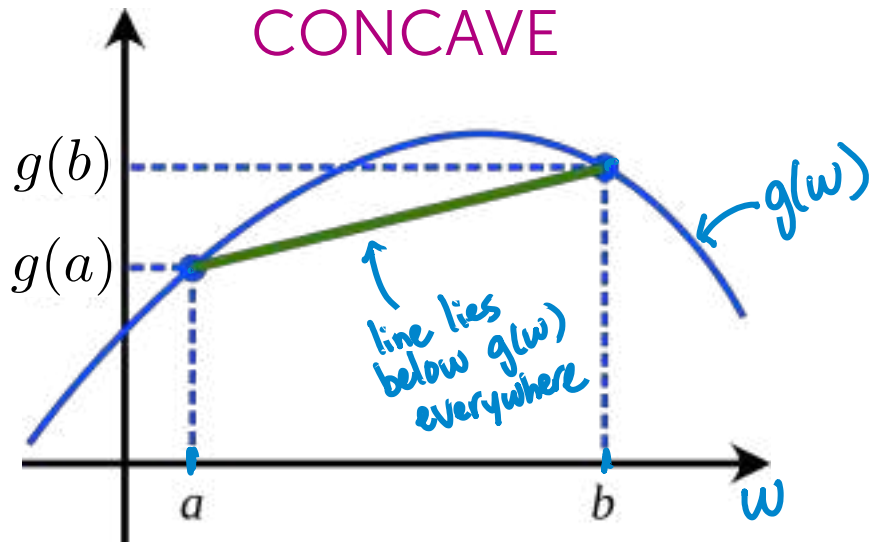
$$\min_{w_0, w_1} \sum_{i=1}^N (y_i - [w_0 + w_1 x_i])^2$$

$\text{RSS}(w_0, w_1)$ is a function
of 2 variables $= g(w_0, w_1)$

An aside on optimization

Convex/concave functions

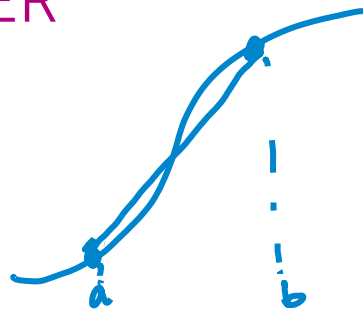
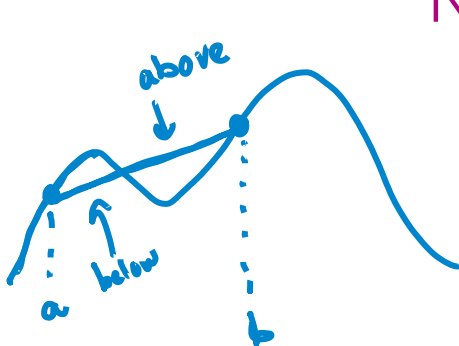
CONCAVE



CONVEX

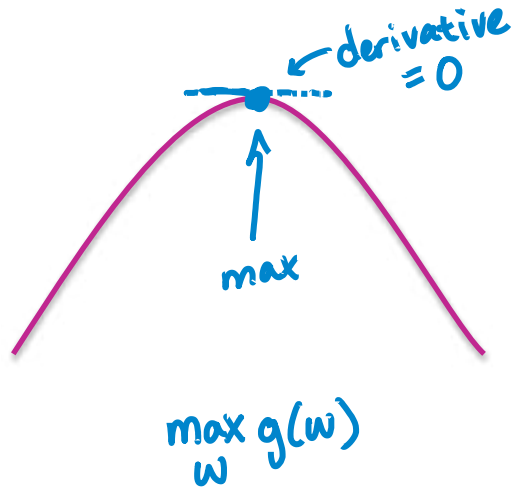


NEITHER

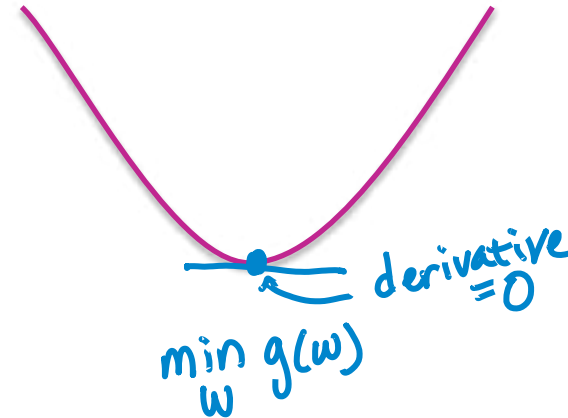


Finding the max or min analytically

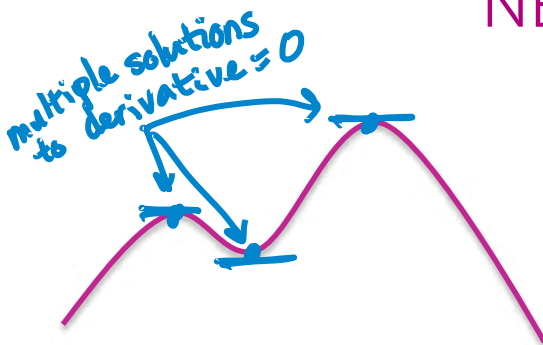
CONCAVE



CONVEX



NEITHER



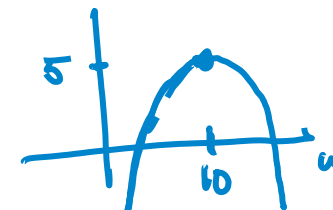
Example:

$$g(w) = 5 - (w - 10)^2$$

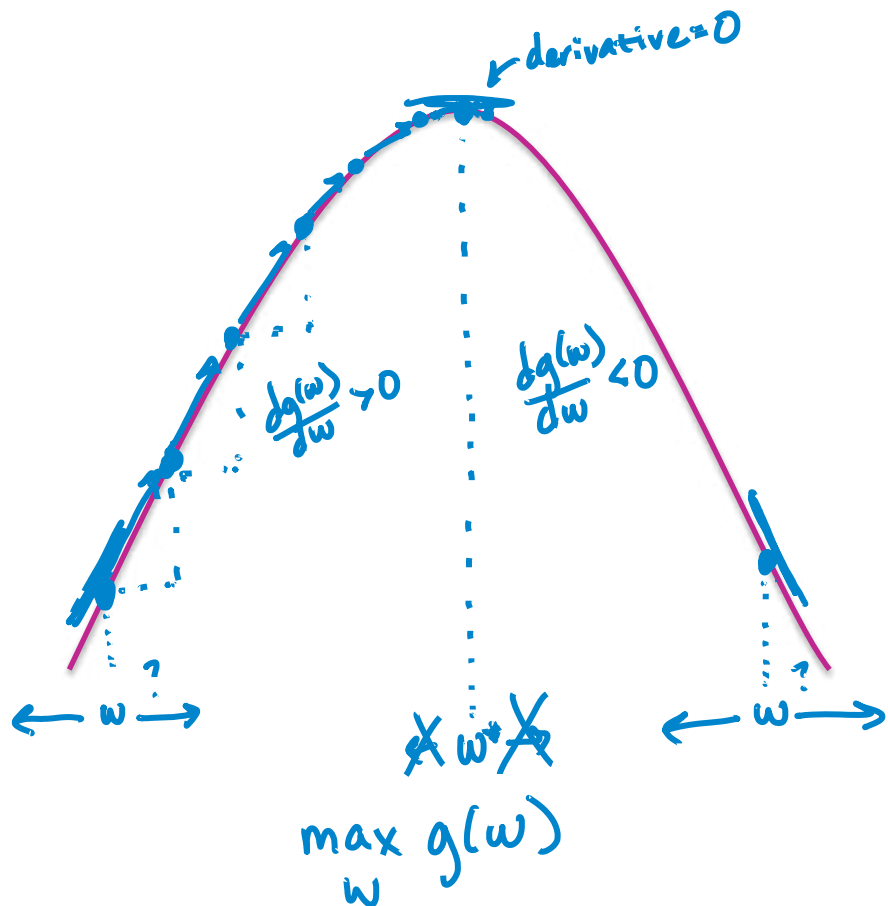
$$\begin{aligned}\frac{dg(w)}{dw} &= 0 - 2(w - 10) \cdot 1 \\ &= -2w + 20\end{aligned}$$

set derivate = 0:

$$\begin{aligned}-2w + 20 &= 0 \\ w &= 10\end{aligned}$$



Finding the max via hill climbing



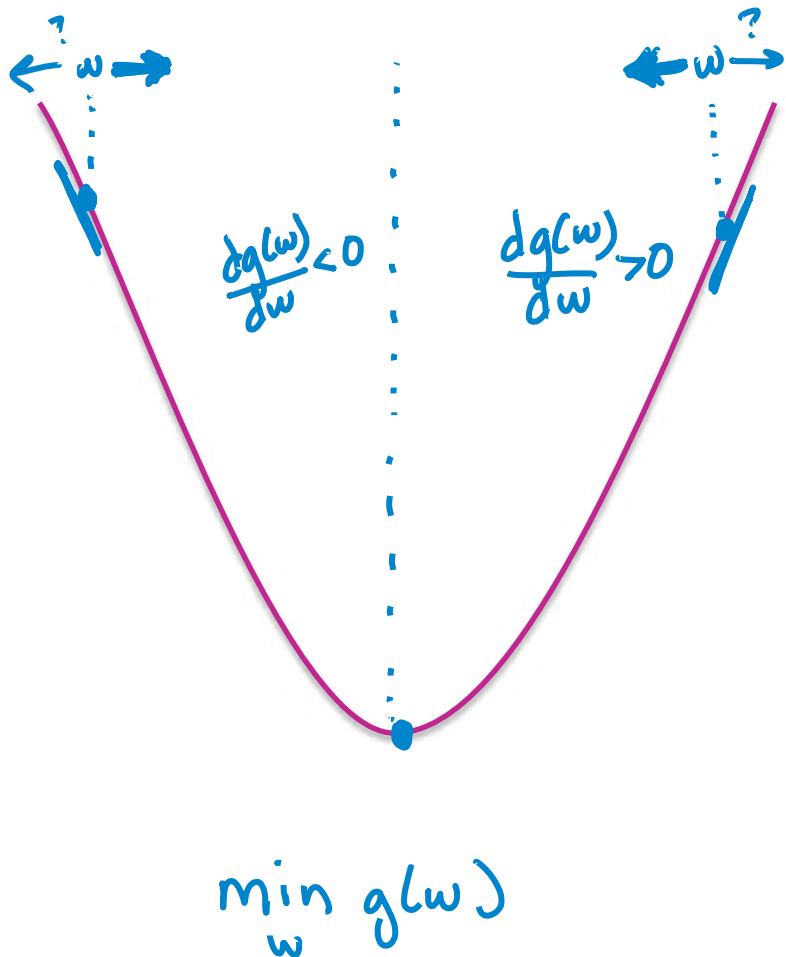
How do we know whether to move w to right or left?
(inc. or dec. the value of w ?)

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} + \eta \frac{dg(w)}{dw}$$

iteration t stepsize

Finding the min via hill descent



When derivative is positive, we
want to decrease w
and when derivative is negative,
we want to increase w

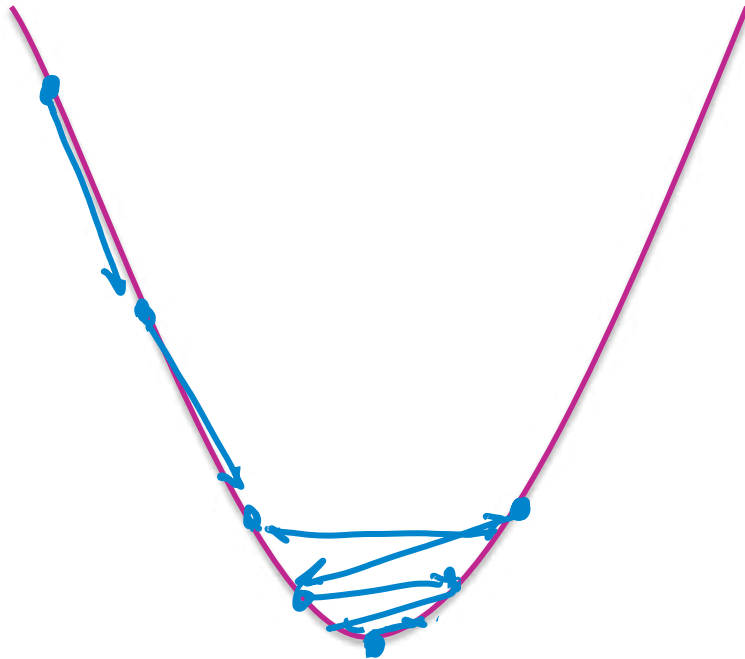
Algorithm:

while not converged
 $w^{(t+1)} \leftarrow w^{(t)} - \eta \left. \frac{dg}{dw} \right|_{w^{(t)}}$

Choosing the stepsize—

Fixed stepsize

η

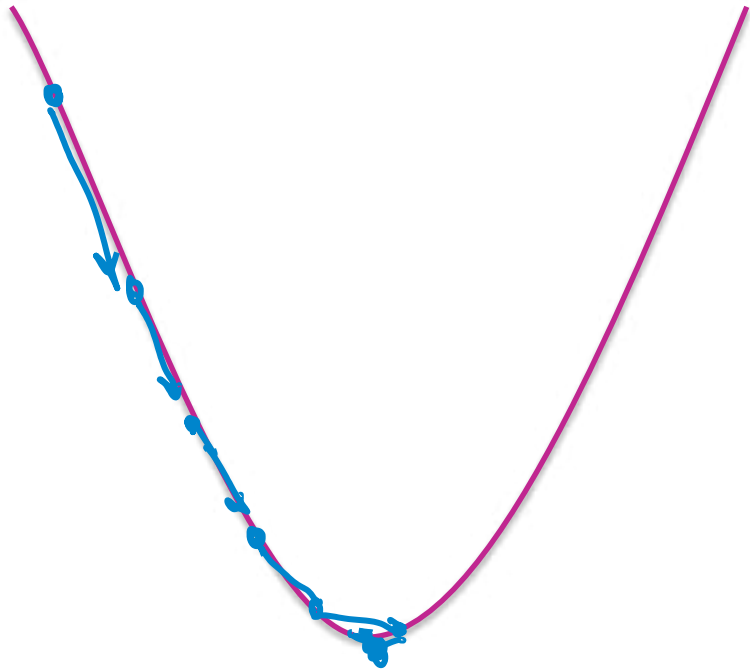


$\eta = 0.1$

Choosing the stepsize—

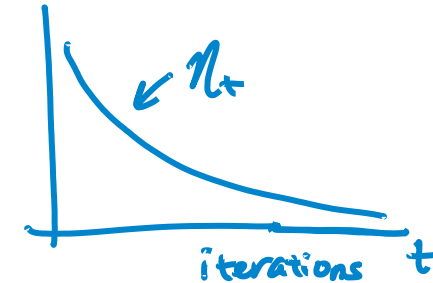
Decreasing stepsize

or stepsize schedule



Common choices:

$$\eta_t = \frac{\alpha}{t}$$
$$\eta_t = \frac{\alpha}{\sqrt{t}}$$



Convergence criteria

For convex functions,
optimum occurs when

$$\frac{dg(w)}{dw} = 0$$

In practice, stop when

$$\left| \frac{dg(w)}{dw} \right| < \epsilon$$

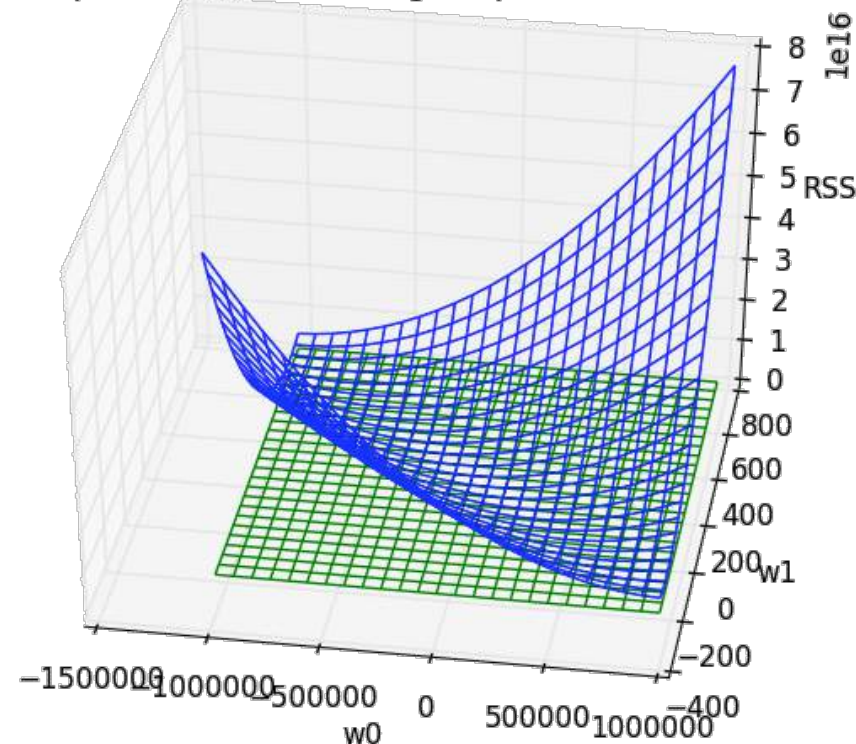
↑ threshold to be set

Algorithm:

while not converged
 $w^{(t+1)} \leftarrow w^{(t)} - \eta \left. \frac{dg}{dw} \right|_{w^{(t)}}$

Moving to multiple dimensions: Gradients

3D plot of RSS with tangent plane at minimum



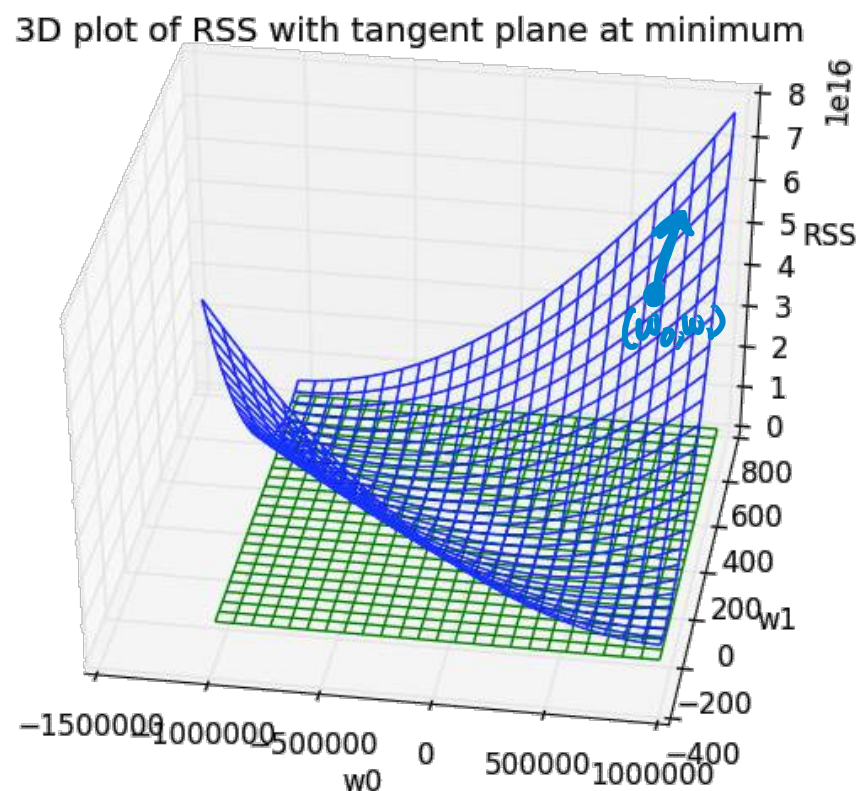
$$\nabla g(\mathbf{w}) = \begin{bmatrix} \frac{\partial g}{\partial w_0} \\ \frac{\partial g}{\partial w_1} \\ \vdots \\ \frac{\partial g}{\partial w_p} \end{bmatrix}$$

gradient $\uparrow$ $[w_0, w_1, \dots, w_p]$

$(p+1)$ -dimensional vector

partial derivative is like a derivative with respect to w_i , treating all other variables as constants

Gradient example



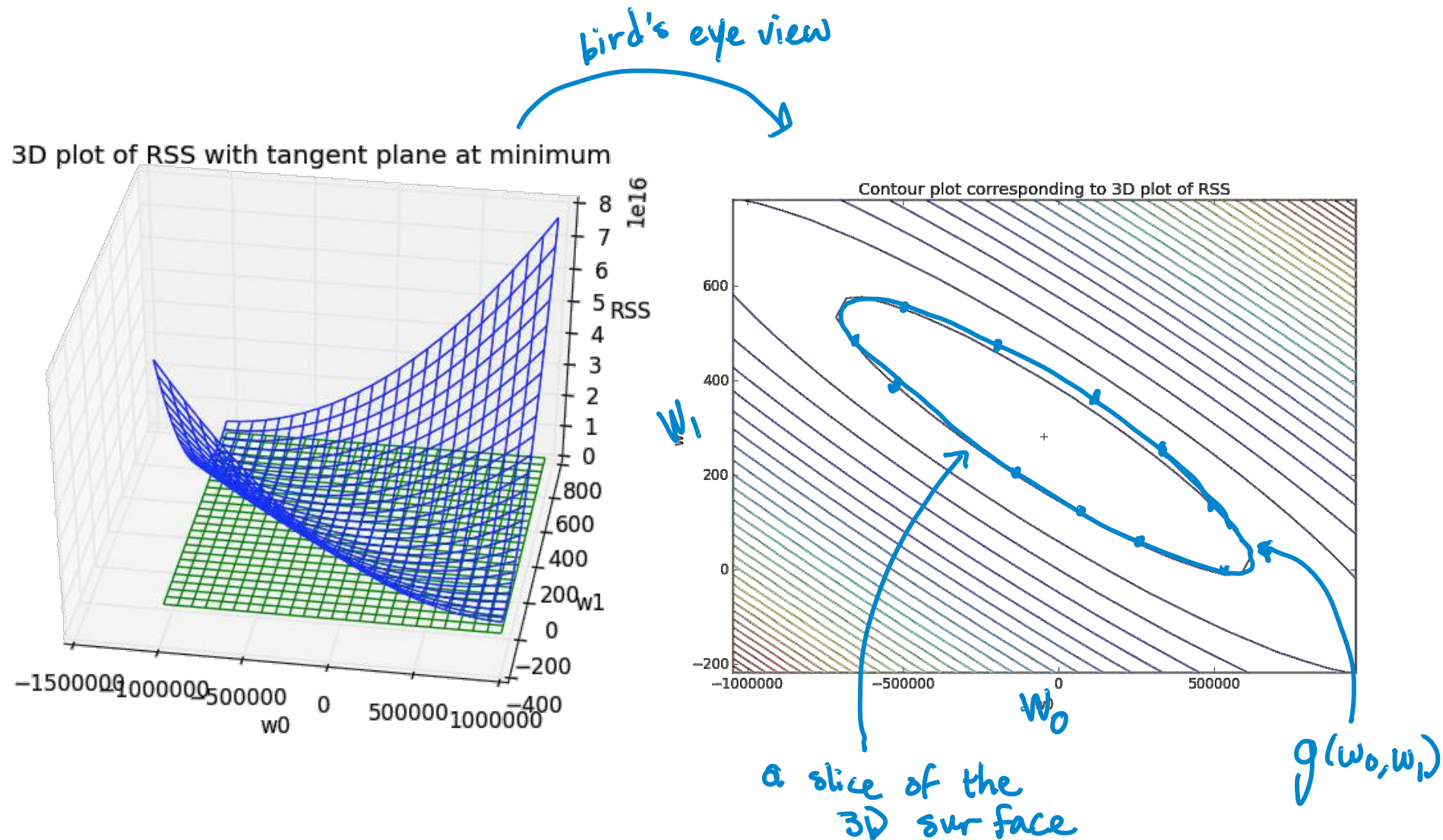
$$g(\mathbf{w}) = 5w_0 + 10w_0w_1 + 2w_1^2$$

$$\frac{\partial g}{\partial w_0} = 5 + 10w_1$$

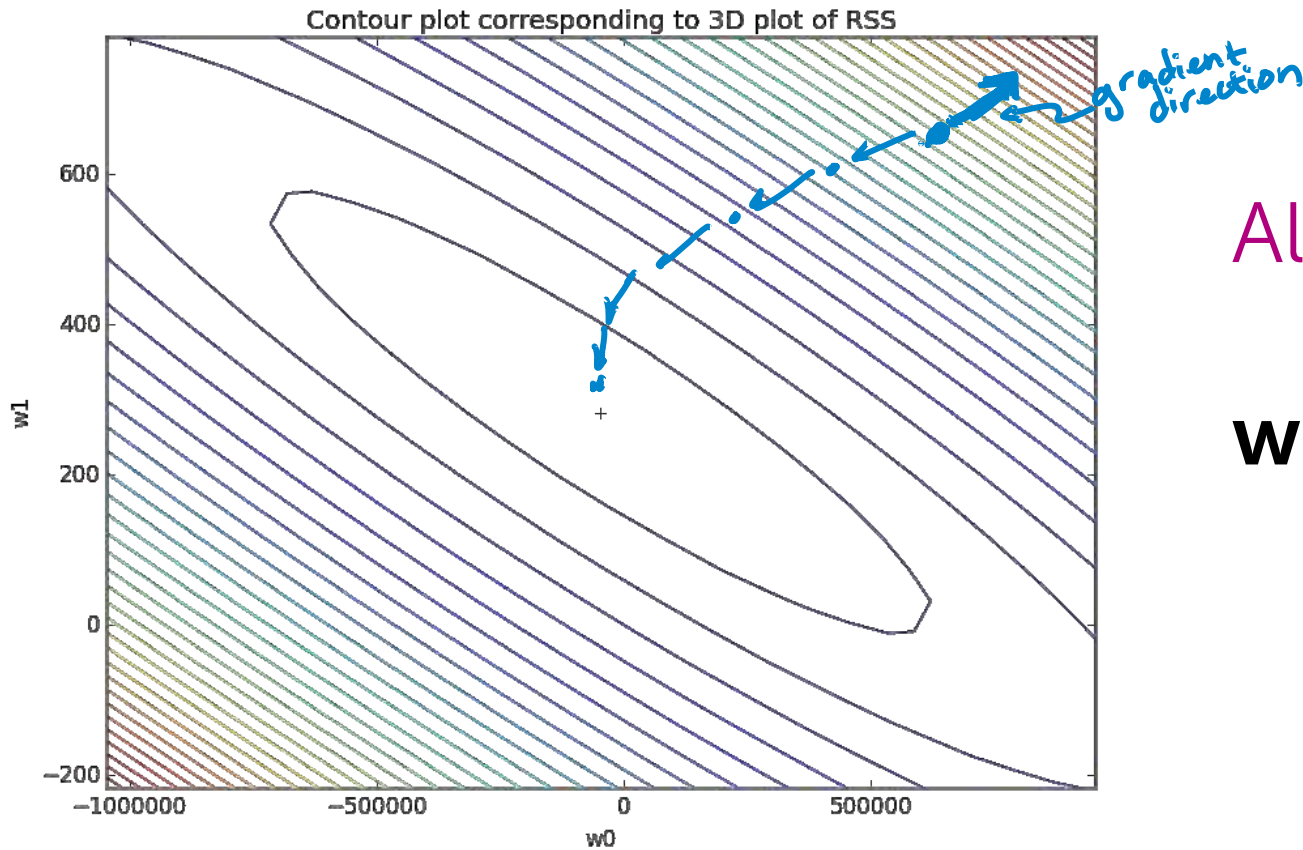
$$\frac{\partial g}{\partial w_1} = 10w_0 + 4w_1$$

$$\nabla g(\mathbf{w}) = \begin{bmatrix} 5 + 10w_1 \\ 10w_0 + 4w_1 \end{bmatrix}$$

Contour plots



Gradient descent



Algorithm:

while not converged

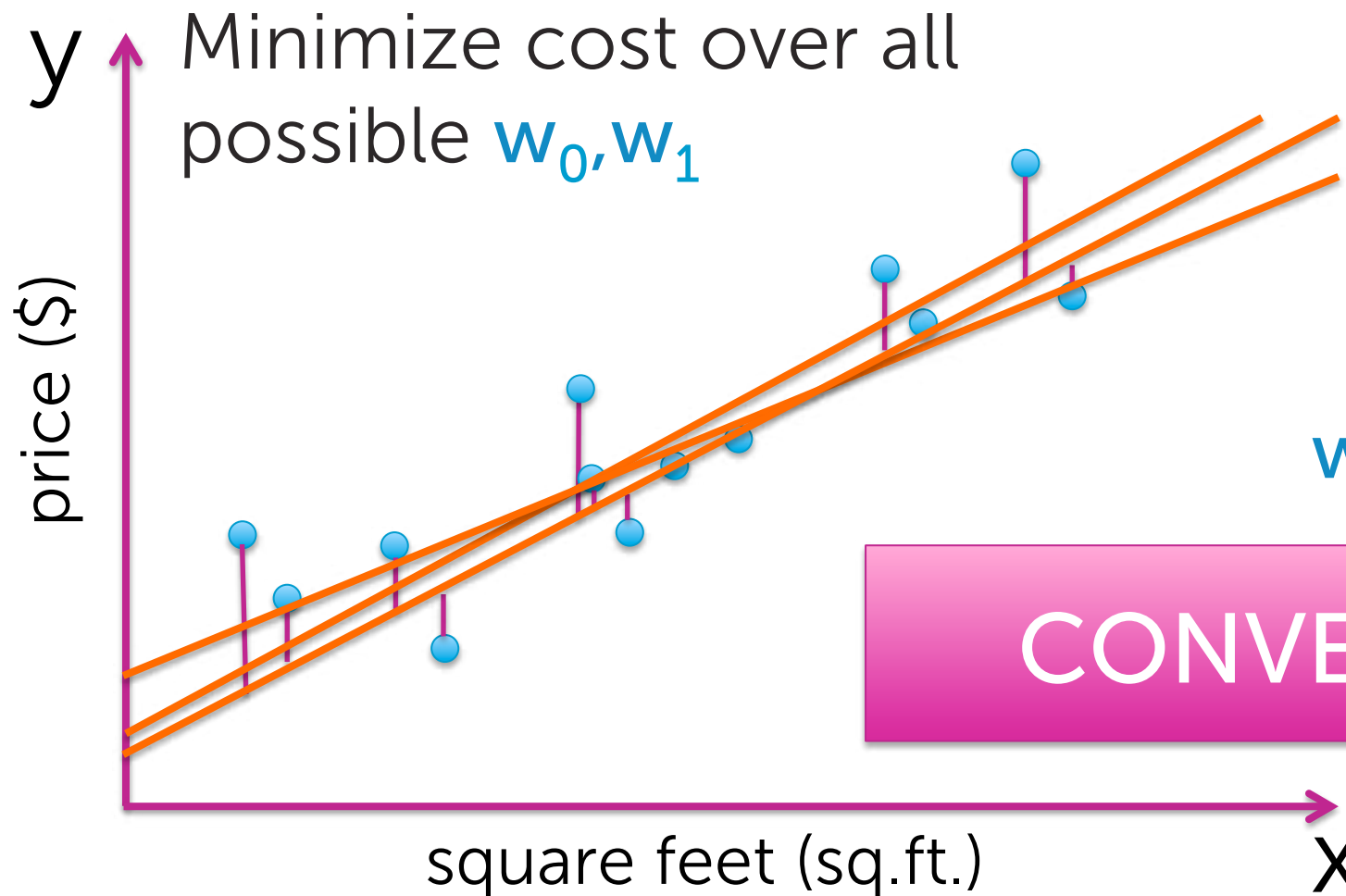
$$w^{(t+1)} \leftarrow w^{(t)} - \eta \nabla g(w^{(t)})$$

$$\begin{bmatrix} \vdots \\ w_0 \\ \vdots \end{bmatrix} \leftarrow \begin{bmatrix} \vdots \\ w_0 \\ \vdots \end{bmatrix} - \eta \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

Convergence:
 $\|\nabla g(w)\| < \epsilon$

Finding the least squares line

Find "best" line



$$\min_{w_0, w_1} \sum_{i=1}^N (y_i - [w_0 + w_1 x_i])^2$$

CONVEX

$\Rightarrow$ solution is unique
+ gradient descent
alg. will converge
to minimum

Compute the gradient

$$\text{RSS}(w_0, w_1) = \sum_{i=1}^N (y_i - [w_0 + w_1 x_i])^2$$

Aside:

$$\begin{aligned} \frac{d}{dw} \sum_{i=1}^N g_i(w) &= \frac{d}{dw} (g_1(w) + g_2(w) + \dots + g_N(w)) \\ &= \frac{d}{dw} g_1(w) + \frac{d}{dw} g_2(w) + \dots + \frac{d}{dw} g_N(w) \\ &= \sum_{i=1}^N \frac{d}{dw} g_i(w) \end{aligned}$$

In our case

$$g_i(w) = (y_i - [w_0 + w_1 x_i])^2$$

$$\frac{\partial \text{RSS}(w)}{\partial w_0} = \sum_{i=1}^N \frac{\partial}{\partial w_0} (y_i - [w_0 + w_1 x_i])^2$$

same for w_1

Compute the gradient

$$\text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])^2$$

Taking the derivative w.r.t. $\mathbf{w}_0$

$$\sum_{i=1}^N 2 (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])' \cdot (-1)$$

$$= -2 \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])$$

Compute the gradient

$$\text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])^2$$

Taking the derivative w.r.t. $\mathbf{w}_1$

$$\begin{aligned} & \sum_{i=1}^N 2(y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])' \cdot (-x_i) \\ &= -2 \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i]) \underline{x_i} \end{aligned}$$

Compute the gradient

$$\text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])^2$$

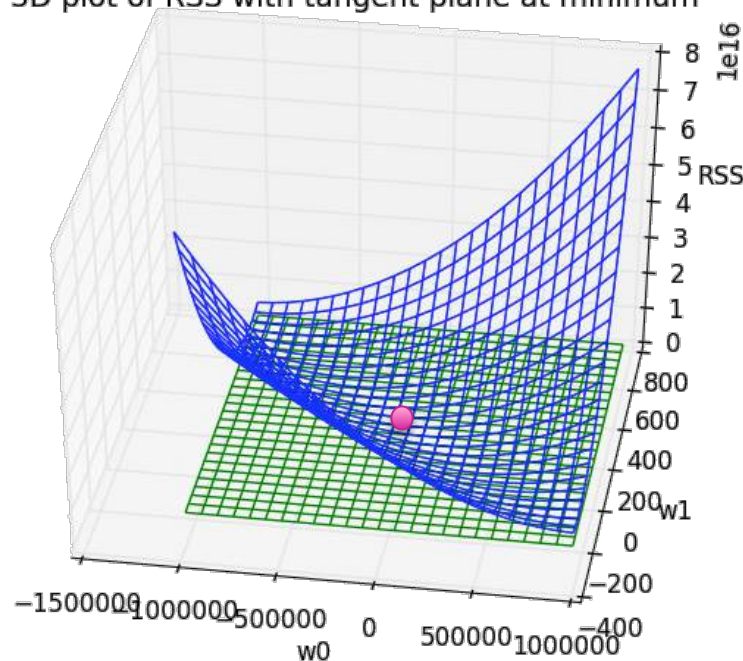
Putting it together:

$$\nabla \text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \begin{bmatrix} -2 \sum_{i=1}^N [y_i - (\mathbf{w}_0 + \mathbf{w}_1 x_i)] \\ -2 \sum_{i=1}^N [y_i - (\mathbf{w}_0 + \mathbf{w}_1 x_i)] x_i \end{bmatrix}$$

Approach 1: Set gradient = 0

$$\nabla \text{RSS}(w_0, w_1) = \begin{bmatrix} -2 \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] \\ -2 \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] x_i \end{bmatrix}$$

3D plot of RSS with tangent plane at minimum



top term:

$$\hat{w}_0 = \frac{\sum_{i=1}^N y_i}{N} - \hat{w}_1 \frac{\sum_{i=1}^N x_i}{N}$$

average sales house price ← $\frac{\sum y_i}{N}$
estimate of the slope ← $\hat{w}_1$
average sq. ft. ← $\frac{\sum x_i}{N}$

bottom term:

$$\sum y_i x_i - \hat{w}_0 \sum x_i - \hat{w}_1 \sum x_i^2 = 0$$

plug in →

$$\hat{w}_1 = \frac{\sum y_i x_i - \frac{\sum y_i \sum x_i}{N}}{\sum x_i^2 - \frac{\sum x_i \sum x_i}{N}}$$

Note:

$$\begin{aligned} &\sum_{i=1}^N y_i \\ &\sum_{i=1}^N x_i \\ &\sum_{i=1}^N y_i x_i \\ &\sum_{i=1}^N x_i^2 \end{aligned}$$

Approach 2: Gradient descent

Interpreting the gradient:

$$\nabla \text{RSS}(w_0, w_1) = \begin{bmatrix} -2 \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] \\ -2 \sum_{i=1}^N [y_i - (w_0 + w_1 x_i)] x_i \end{bmatrix} = \begin{bmatrix} -2 \sum_{i=1}^N [y_i - \hat{y}_i(w_0, w_1)] \\ -2 \sum_{i=1}^N [y_i - \hat{y}_i(w_0, w_1)] x_i \end{bmatrix}$$

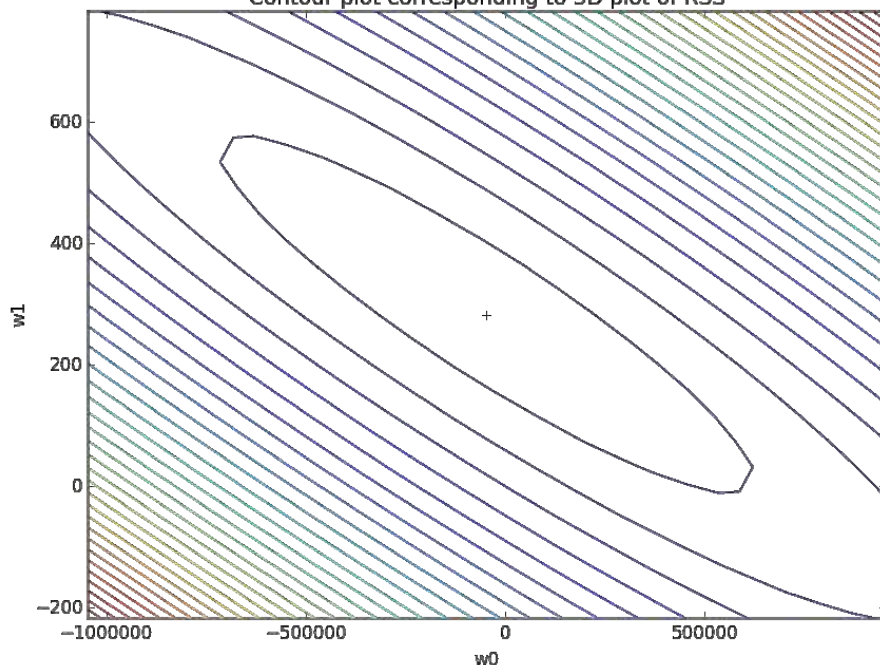
Handwritten annotations:

- y_i : actual house sales observation
- $(w_0 + w_1 x_i)$: predicted value $\hat{y}_i(w_0, w_1)$

Approach 2: Gradient descent

$$\nabla \text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \begin{bmatrix} -2 \sum_{i=1}^N [y_i - \hat{y}_i(\mathbf{w}_0, \mathbf{w}_1)] \\ -2 \sum_{i=1}^N [y_i - \hat{y}_i(\mathbf{w}_0, \mathbf{w}_1)] x_i \end{bmatrix}$$

Contour plot corresponding to 3D plot of RSS



while not converged $(-2) \cdot (-n)$

$$\begin{bmatrix} w_0^{(t+1)} \\ w_1^{(t+1)} \end{bmatrix} \leftarrow \begin{bmatrix} w_0^{(t)} \\ w_1^{(t)} \end{bmatrix} + 2\eta \begin{bmatrix} \sum_{i=1}^N [y_i - \hat{y}_i(w_0^{(t)}, w_1^{(t)})] \\ \sum_{i=1}^N [y_i - \hat{y}_i(w_0^{(t)}, w_1^{(t)})] x_i \end{bmatrix}$$

If overall, under predicting $\hat{y}_i$, then $\sum [y_i - \hat{y}_i]$ is positive

→ w_0 is going to increase

similar intuition for w_1 , but multiply by x_i

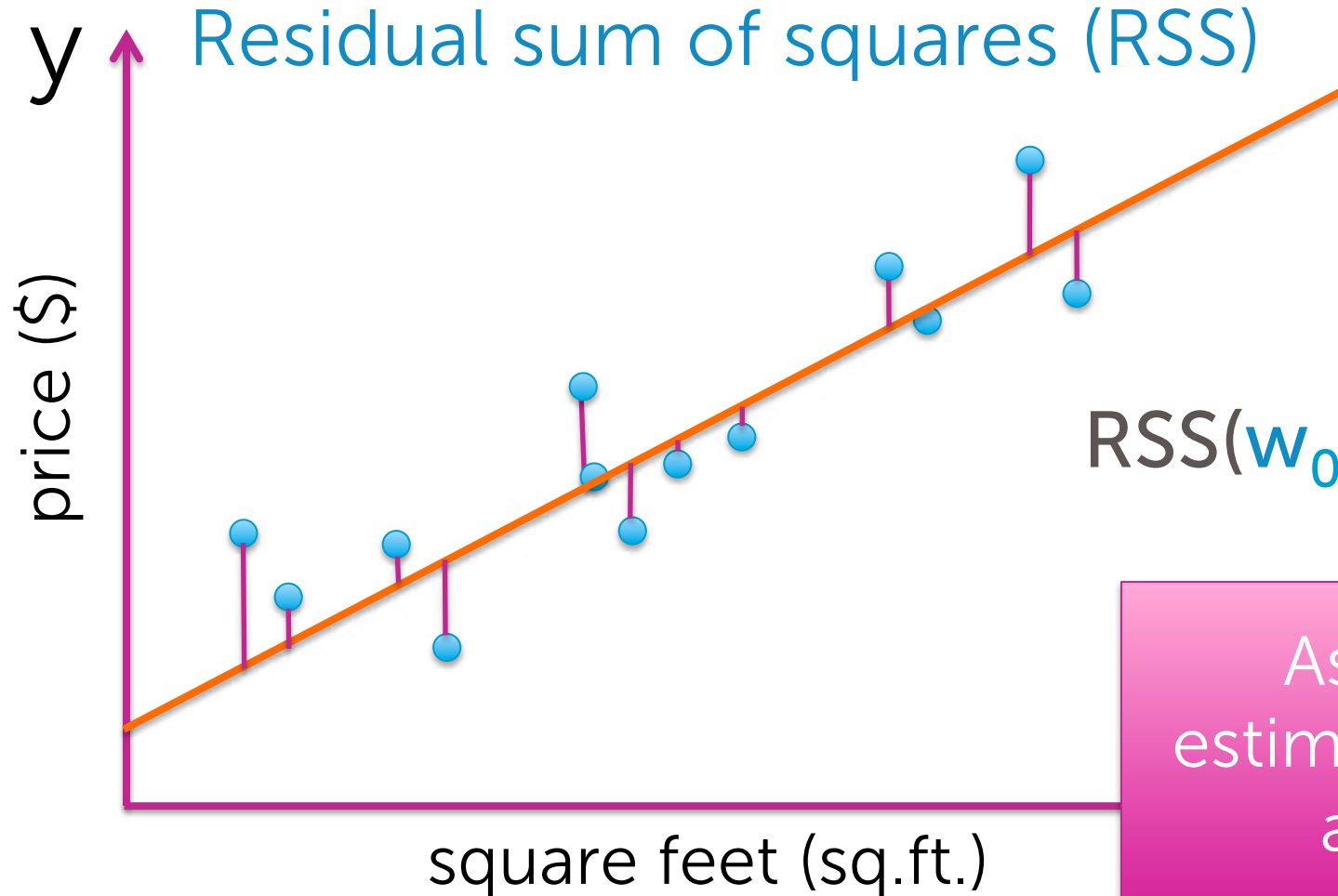
Comparing the approaches

- For most ML problems, cannot solve $\text{gradient} = 0$
- Even if solving $\text{gradient} = 0$ is feasible, gradient descent can be more efficient
- Gradient descent relies on choosing stepsize and convergence criteria

Influence of high leverage points

Asymmetric errors

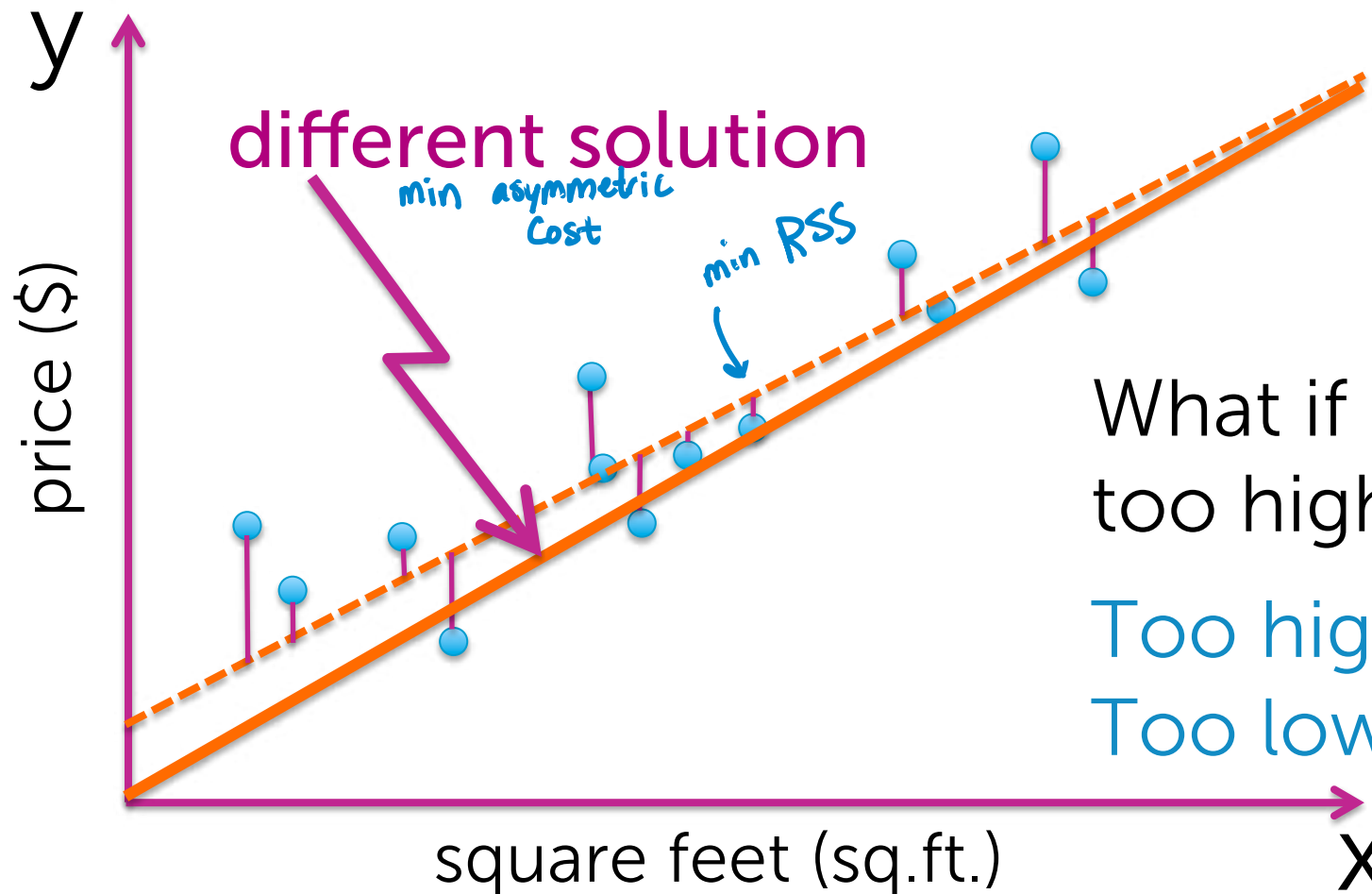
Symmetric cost functions



$$\text{RSS}(\mathbf{w}_0, \mathbf{w}_1) = \sum_{i=1}^N (y_i - [\mathbf{w}_0 + \mathbf{w}_1 x_i])^2$$

Assumes cost of over-estimating sales price is same as under-estimating

Asymmetric cost functions



What if cost of listing house too high has **bigger cost**?

Too high $\rightarrow$ no offers ($\$=0$)

Too low $\rightarrow$ offers for lower \$

Summary for simple linear regression

What you can do now...

- Describe the input (features) and output (real-valued predictions) of a regression model
- Calculate a goodness-of-fit metric (e.g., RSS)
- Estimate model parameters to minimize RSS using gradient descent
- Interpret estimated model parameters
- Exploit the estimated model to form predictions
- Discuss the possible influence of high leverage points
- Describe intuitively how fitted line might change when assuming different goodness-of-fit metrics